Appendix A Proof of proposition 1

The overall profit of digital supply chain under the collaboration orientation can be written as:

$$R_T=\int_0^{+\infty}e^{-\tau t}\left\{(\pi_M+\pi_R)D(t)-\frac{1}{2}\eta_S E_S^2(t)-\frac{1}{2}\eta_P E_P^2(t)-\frac{1}{2}\eta_R E_R^2(t)+k(t)(S(t)-\overline{S})-k(t)(\overline{P}-P(t))\right\}dt \quad \text{EqA1}$$

The optimal value function of the overall digital supply chain at t-moment can be presented as:

$$\mathrm{V}_T(S,P)=\max_{E_S(t),E_P(t),E_R(t)}\int_t^{+\infty}e^{-\rho(\tau-t)}\left\{(\pi_M+\pi_R)D(t)-\frac{1}{2}\eta_S E_S^2(t)-\frac{1}{2}\eta_P E_P^2(t)-\frac{1}{2}\eta_R E_R^2(t)+k(t)(S(t)-\overline{S})-k(t)(\overline{P}-P(t))\right\}dt \quad \text{EqA2}$$

Above two function have different inflexion points in calculating the discounted value. According to optimal control theory, their relationship can be presented as: $R_T=e^{-\rho t}\mathrm{V}_T(S,P)$

In order to obtain the Markovian perfect equilibrium solution of the cooperative game, we assume that retailer, manufacturer and government have a continuous bounded differential profit function satisfying the Hamilton-Jacobi-Bellman (HJB) equation for any variable X⩾0. For any S⩾0, P⩾0, $\mathrm{V}_T(S,P)$ satisfies the HJB condition as follows:

$$\rho\mathrm{V}_T(S,P)=\max_{E_S(t),E_P(t),E_R(t)}\left\{\begin{aligned}&(\pi_M+\pi_R)(D_0+d_1S(t)+d_2P(t)+d_3E_R(t))-\frac{1}{2}\eta_S E_S^2(t)-\frac{1}{2}\eta_P E_P^2(t)-\frac{1}{2}\eta_R E_R^2(t)+k(t)(S(t)-\overline{S})-k(t)(\overline{P}-P(t))\\&+V_{TS}^{'}\left[s_1E_S(t)-\delta S(t)\right]+V_{TP}^{'}\left[p_1E_P(t)+p_2S(t)-\omega P(t)\right]\end{aligned}\right\}dt \quad \text{EqA3}$$

The optimal value strategy satisfies that the first-order derivative is zero and the second-order derivative is negative. Intuitively, the second-order derivative is negative, which indicates that $\rho\mathrm{V}_T(S,P)$ is concave on $E_S(t),E_P(t)$ and $E_R(t)$. The Hessian matrix of $E_S(t),E_P(t)$ and $E_R(t)$ is H= $\begin{bmatrix}-\eta_S & 0 & 0\\ 0 & -\eta_P & 0\\ 0 & 0 & -\eta_R\end{bmatrix}$ <0, the optimal strategy for the global maximum can be calculated by solving the following equation:

$$\begin{cases} \dfrac{\partial \rho V_T(X,S)}{\partial E_S(t)} = -\eta_S E_S(t) + s_1 V_{TS}^{'} = 0 \\ \dfrac{\partial \rho V_T(X,S)}{\partial E_P(t)} = -\eta_P E_P(t) + p_1 V_{TP}^{'} = 0 \\ \dfrac{\partial \rho V_T(X,S)}{\partial E_R(t)} = -\eta_R E_R(t) + d_3\left(\pi_M + \pi_R\right) = 0 \end{cases} \quad \text{EqA4}$$

which can be calculated:

$$\begin{cases} E_S^*(t) = \dfrac{s_1 V_{TS}^{'}}{\eta_S} \\ E_P^*(t) = \dfrac{p_1 V_{TP}^{'}}{\eta_P} \\ E_R^*(t) = \dfrac{d_3\left(\pi_M + \pi_R\right)}{\eta_R} \end{cases} \quad \text{EqA5}$$

Substitute EqA5 into EqA3 and simplify, we get

$$\rho \mathrm{V}_T(S,P) = \begin{Bmatrix} (\pi_M + \pi_R)D_0 + ((\pi_M + \pi_R)d_1 + k(t) - \delta V_{TS}^{'} + p_2 V_{TP}^{'})S(t) + ((\pi_M + \pi_R)d_2 + k(t) - \omega V_{TP}^{'})P(t) + \dfrac{1}{2}\dfrac{(d_3(\pi_M + \pi_R))^2}{\eta_R} \\ + \dfrac{1}{2}\dfrac{(s_1 V_{TS}^{'})^2}{\eta_S} + \dfrac{1}{2}\dfrac{(p_1 V_{TP}^{'})^2}{\eta_P} - k(t)(\overline{S} + \overline{P}) \end{Bmatrix} dt \quad \text{EqA6}$$

Since $\mathrm{V}_T(S,P)$ is the linear function of $S(t)$ and $P(t)$, we assume that $V_T(S,P) = a_1 S(t) + b_1 P(t) + c_1$, then $V_{TS}^{'} = a_1, V_{TP}^{'} = b_1$, where a_1, b_1, c_1 are constants. Substitute $V_T(S,P)$ and its first partial derivative into EqA6 and simplify, the following equation can be obtained:

$$\rho\left[a_1 S(t)+b_1 P(t)+c_1\right]=((\pi_M+\pi_R)d_1+k(t)-\delta a_1+p_2 b_1)S(t)+((\pi_M+\pi_R)d_2+k(t)-\omega b_1)P(t)+\frac{1}{2}\frac{(d_3(\pi_M+\pi_R))^2}{\eta_R}$$
$$+\frac{1}{2}\frac{(s_1 a_1)^2}{\eta_S}+\frac{1}{2}\frac{(p_1 b_1)^2}{\eta_P}+(\pi_M+\pi_R)D_0-k(t)(\overline{S}+\overline{P}) \quad \text{EqA7}$$

Using the method of undetermined coefficients, we can get the following ternary linear functions:

$$\begin{cases} \rho a_1=(\pi_M+\pi_R)d_1+k(t)-\delta a_1+p_2 b_1 \\ \rho b_1=(\pi_M+\pi_R)d_2+k(t)-\omega b_1 \\ \rho c_1=\frac{1}{2}\frac{(d_3(\pi_M+\pi_R))^2}{\eta_R}+\frac{1}{2}\frac{(s_1 a_1)^2}{\eta_S}+\frac{1}{2}\frac{(p_1 b_1)^2}{\eta_P}+(\pi_M+\pi_R)D_0-k(t)(\overline{S}+\overline{P}) \end{cases} \quad \text{EqA8}$$

the parameter values of a_1, b_1, c_1 can be solved as:

$$\begin{cases} a_1=\frac{\left((\pi_M+\pi_R)d_1+k(t)\right)(\rho+\omega)+p_2\left((\pi_M+\pi_R)d_2+k(t)\right)}{(\rho+\delta)(\rho+\omega)} \\ b_1=\frac{(\pi_M+\pi_R)d_2+k(t)}{\rho+\omega} \\ c_1=\frac{1}{2}\frac{(d_3(\pi_M+\pi_R))^2}{\rho\eta_R}+\frac{1}{2}\frac{s_1^2((\pi_M+\pi_R)(d_1\rho+d_1\omega+d_2 p_2)+k(t)(\rho+\omega+p_2))^2}{\rho\eta_S(\rho+\delta)^2(\rho+\omega)^2}+\frac{1}{2}\frac{p_1^2((\pi_M+\pi_R)d_2+k(t))^2}{\rho\eta_P(\rho+\omega)^2}+\frac{(\pi_M+\pi_R)D_0}{\rho}-\frac{k(t)(\overline{S}+\overline{P})}{\rho} \end{cases} \quad \text{EqA9}$$

As a result, $V_T(S,P)=e^{-\rho t}(a_1 S(t)+b_1 P(t)+c_1)$, the optimal strategy for the global maximum can be calculated respectively as follow:

$$\begin{cases} E_S^*(t) = \dfrac{s_1((\pi_M + \pi_R)(d_1\rho + d_1\omega + d_2 p_2) + k(t)(\rho + \omega + p_2))}{\eta_S(\rho+\delta)(\rho+\omega)} \\ E_P^*(t) = \dfrac{p_1((\pi_M + \pi_R)d_2 + k(t))}{\eta_P(\rho+\omega)} \\ E_R^*(t) = \dfrac{d_3(\pi_M + \pi_R)}{\eta_R} \end{cases} \quad \text{EqA10}$$

Then, the government make decision on optimal policy strategy $k(t)$ to maximize its own profit, and the optimal value function of government at t-moment can be presented as:

$$V_G(S,P) = \max_{k(t)} \int_0^{+\infty} e^{-\rho(\tau-t)} \left\{ g_s S(t) + g_p P(t) - \frac{1}{2}\eta_k k^2(\mathrm{t}) \right\} dt \quad \text{EqA11}$$

For any S⩾0, P⩾0, $\mathrm{V}_G(S,P)$ satisfies the HJB condition as follows:

$$\rho \mathrm{V}_G(S,P) = \max_{k(t)} \left\{ g_s S(t) + g_p P(t) - \frac{1}{2}\eta_k k^2(\mathrm{t}) + V_{GS}^{'}\left[s_1 E_S(t) - \delta S(t) \right] + V_{GP}^{'}\left[p_1 E_P(t) + p_2 S(t) - \omega P(t) \right] \right\} dt \quad \text{EqA12}$$

Substitute $E_S(t), E_P(t)$ and $E_R(t)$ into EqA12, we can obtain:

$$\rho V_G(S,P) = \max_{k(t)} \int_t^{+\infty} e^{-\rho(\tau-t)} \left\{ \begin{array}{l} g_s S(t) + g_p P(t) - \dfrac{1}{2}\eta_k k^2(\mathrm{t}) + V_{GS}^{'}\left[\dfrac{s_1^{\,2}((\pi_M + \pi_R)(d_1\rho + d_1\omega + d_2 p_2) + k(t)(\rho + \omega + p_2))}{\eta_S(\rho+\delta)(\rho+\omega)} - \delta S(t) \right] \\ + V_{GP}^{'}\left[\dfrac{p_1^{\,2}((\pi_M + \pi_R)d_2 + k(t))}{\eta_P(\rho+\omega)} + p_2 S(t) - \omega P(t) \right] \end{array} \right\} dt \quad \text{EqA13}$$

Simplify EqA13 to a function with respect to government strategy k(t) can obtain:

$$\rho V_G(S,P)=\max_{k(t)}\int_t^{+\infty}e^{-\rho(\tau-t)}\left\{\begin{aligned}&\left(-\frac{1}{2}\eta_k\right)k^2(t)+\left(\frac{V_{GS}^{'}s_1^{\,2}(\rho+\omega+p_2)}{\eta_S(\rho+\delta)(\rho+\omega)}+\frac{V_{GP}^{'}p_1^{\,2}}{\eta_P(\rho+\omega)}\right)k(t)+g_sS(t)+g_pP(t)+\frac{V_{GS}^{'}s_1^{\,2}(\pi_M+\pi_R)(d_1\rho+d_1\omega+d_2p_2)}{\eta_S(\rho+\delta)(\rho+\omega)}-V_{GS}^{'}\delta S(t)\\&+\frac{V_{GP}^{'}p_1^{\,2}(\pi_M+\pi_R)d_2}{\eta_P(\rho+\omega)}+V_{GP}^{'}p_2S(t)-V_{GP}^{'}\omega P(t)\end{aligned}\right\}dt$$

EqA14

Intuitively, the second-order derivative is negative, which indicates that $\rho V_G(S,P)$ is concave on $k(t)$. The necessary condition of optimal value strategy requires $\frac{\partial\rho V_G(S,P)}{\partial k(t)}=0$. By solving this equation, the optimal strategy $k^*(t)$ for the global maximum can be calculated.

$$k^*(t)=\frac{\eta_P V_{GS}^{'}s_1^{\,2}(\rho+\omega+p_2)+\eta_S(\rho+\delta)V_{GP}^{'}p_1^{\,2}}{\eta_K\eta_P\eta_S(\rho+\delta)(\rho+\omega)} \qquad \text{EqA15}$$

Since $V_G(S,P)$ is the linear function of $S(t)$ and $P(t)$, we assume that $V_G(S,P)=AS(t)+BP(t)+C$, then $V_{GS}^{'}=A, V_{GP}^{'}=B$, where A,B,C are constants. Substitute $V_G(S,P)$ and its first partial derivative into EqA14 and simplify, the following equation can be obtained:

$$\rho\left[AS(t)+BP(t)+C\right]=\left(g_s-\delta V_{GS}^{'}+p_2V_{GP}^{'}\right)S(t)+\left(g_p-\omega V_{GP}^{'}\right)P(t)-\frac{1}{2}\eta_k k^2(\mathrm{t})+\frac{V_{GS}^{'}s_1^{\,2}((\pi_M+\pi_R)(d_1\rho+d_1\omega+d_2p_2)+k(t)(\rho+\omega+p_2))}{\eta_S(\rho+\delta)(\rho+\omega)}+\frac{V_{GP}^{'}p_1^{\,2}((\pi_M+\pi_R)d_2+k(t))}{\eta_P(\rho+\omega)}$$

EqA16

Using the method of undetermined coefficients, we can get the following ternary linear functions:

$$\begin{cases} \rho A = g_s - \delta A + p_2 B \\ \rho B = g_p - \omega B \\ \rho C = \dfrac{A s_1^2((\pi_M+\pi_R)(d_1\rho + d_1\omega + d_2 p_2) + k(t)(\rho+\omega+p_2))}{\eta_S(\rho+\delta)(\rho+\omega)} + \dfrac{B p_1^2((\pi_M+\pi_R)d_2 + k(t))}{\eta_P(\rho+\omega)} - \dfrac{1}{2}\eta_k k^2(\mathrm{t}) \end{cases} \quad \text{EqA17}$$

the parameter values of A, B, C can be solved as:

$$\begin{cases} A = \dfrac{(\rho+\omega)g_S + p_2 g_p}{(\rho+\delta)(\rho+\omega)} \\ B = \dfrac{g_p}{\rho+\omega} \\ C = \dfrac{((\rho+\omega)g_S + p_2 g_p)s_1^2((\pi_M+\pi_R)(d_1\rho+d_1\omega+d_2p_2)+k(t)(\rho+\omega+p_2))}{\rho(\rho+\delta)^2(\rho+\omega)^2\eta_S} + \dfrac{g_p p_1^2((\pi_M+\pi_R)d_2+k(t))}{\rho(\rho+\omega)^2\eta_P} - \dfrac{\eta_k k^2(\mathrm{t})}{2\rho} \end{cases} \quad \text{EqA18}$$

The optimal strategy of the government, retailer and the manufacturer are as below:

$$k^*(t) = \frac{((\rho+\omega)g_S + p_2 g_p)\eta_P s_1^2(\rho+\omega+p_2) + \eta_S g_p p_1^2(\rho+\delta)^2}{(\rho+\delta)^2(\rho+\omega)^2\eta_K\eta_P\eta_S}$$

$$\begin{cases} E_S^*(t) = \dfrac{s_1(\pi_M+\pi_R)(d_1\rho+d_1\omega+d_2p_2)}{\eta_S(\rho+\delta)(\rho+\omega)} + \dfrac{s_1^3\eta_P(\rho+\omega+p_2)^2((\rho+\omega)g_S+p_2g_p)}{(\rho+\delta)^3(\rho+\omega)^3\eta_K\eta_P\eta_S^2} + \dfrac{s_1(\rho+\omega+p_2)\eta_S g_p p_1^2(\rho+\delta)^2}{(\rho+\delta)^3(\rho+\omega)^3\eta_K\eta_P\eta_S^2} \\ E_P^*(t) = \dfrac{p_1(\pi_M+\pi_R)d_2}{\eta_P(\rho+\omega)} + \dfrac{p_1(((\rho+\omega)g_S+p_2g_p)\eta_P s_1^2(\rho+\omega+p_2)+\eta_S(\rho+\delta)^2 g_p p_1^2)}{(\rho+\delta)^2(\rho+\omega)^3\eta_K\eta_P^2\eta_S} \\ E_R^*(t) = \dfrac{d_3(\pi_M+\pi_R)}{\eta_R} \end{cases} \quad \text{EqA19}$$

Substitute $E_S(t), E_P(t)$ and $E_R(t)$ into $\dot{S}(t)$ and $\dot{P}(t)$ and, we can obtain:

$$\begin{cases} \dot{S}(t) = \dfrac{s_1^{\ 2}(\pi_M + \pi_R)(d_1\rho + d_1\omega + d_2 p_2)}{\eta_S(\rho+\delta)(\rho+\omega)} + \dfrac{s_1^{\ 4}\eta_P(\rho+\omega+p_2)^2((\rho+\omega)g_S + p_2 g_p)}{(\rho+\delta)^3(\rho+\omega)^3\eta_K\eta_P\eta_S^{\ 2}} + \dfrac{s_1^{\ 2}(\rho+\omega+p_2)\eta_S g_p p_1^{\ 2}(\rho+\delta)^2}{(\rho+\delta)^3(\rho+\omega)^3\eta_K\eta_P\eta_S^{\ 2}} - \delta S(t) \\ \dot{P}(t) = \dfrac{p_1^{\ 2}(\pi_M+\pi_R)d_2}{\eta_P(\rho+\omega)} + \dfrac{p_1^{\ 2}(((\rho+\omega)g_S + p_2 g_p)\eta_P s_1^{\ 2}(\rho+\omega+p_2) + \eta_S(\rho+\delta)^2 g_p p_1^{\ 2})}{(\rho+\delta)^2(\rho+\omega)^3\eta_K\eta_P^{\ 2}\eta_S} + p_2 S(t) - \omega P(t) \end{cases} \quad \text{EqA20}$$

The state of trajectory of the digital level can be obtained by solving the first-order differential equation EqA20, and the it can be presented as:

$S^*(t) = S_\infty + (S_0 - S_\infty)e^{-\delta t}$, where

$$S_\infty = \frac{s_1^{\ 2}(\pi_M + \pi_R)(d_1\rho + d_1\omega + d_2 p_2)}{\delta\eta_S(\rho+\delta)(\rho+\omega)} + \frac{s_1^{\ 4}\eta_P(\rho+\omega+p_2)^2((\rho+\omega)g_S + p_2 g_p)}{\delta(\rho+\delta)^3(\rho+\omega)^3\eta_K\eta_P\eta_S^{\ 2}} + \frac{s_1^{\ 2}(\rho+\omega+p_2)\eta_S g_p p_1^{\ 2}(\rho+\delta)^2}{\delta(\rho+\delta)^3(\rho+\omega)^3\eta_K\eta_P\eta_S^{\ 2}} \quad \text{EqA21}$$

Substitute $S^*(t)$ into $\dot{P}(t)$, we can obtain that

$$\dot{P}(t) = \frac{p_1^{\ 2}(\pi_M+\pi_R)d_2}{\eta_P(\rho+\omega)} + \frac{p_1^{\ 2}(((\rho+\omega)g_S + p_2 g_p)\eta_P s_1^{\ 2}(\rho+\omega+p_2) + \eta_S(\rho+\delta)^2 g_p p_1^{\ 2})}{(\rho+\delta)^2(\rho+\omega)^3\eta_K\eta_P^{\ 2}\eta_S} + p_2(S_\infty + (S_0 - S_\infty)e^{-\delta t}) - \omega P(t) \quad \text{EqA22}$$

The state of trajectory of the carbon adatement level can be obtained by solving the first-order differential equation EqA20 and the it can be presented as:

$P^*(t) = P_\infty + \left(P_0 - P_\infty - \dfrac{p_2(S_0 - S_\infty)}{\omega - \delta}\right)e^{-\omega t} + \dfrac{p_2(S_0 - S_\infty)}{\omega-\delta}e^{-\delta t}$, where

$$\begin{aligned} P_\infty = {} & \frac{p_1^{\ 2}(\pi_M+\pi_R)d_2}{\omega\eta_P(\rho+\omega)} + \frac{p_1^{\ 2}(((\rho+\omega)g_S + p_2 g_p)\eta_P s_1^{\ 2}(\rho+\omega+p_2) + \eta_S(\rho+\delta)^2 g_p p_1^{\ 2})}{\omega(\rho+\delta)^2(\rho+\omega)^3\eta_K\eta_P^{\ 2}\eta_S} \\ & \frac{p_2 s_1^{\ 2}(\pi_M + \pi_R)(d_1\rho + d_1\omega + d_2 p_2)}{\omega\delta\eta_S(\rho+\delta)(\rho+\omega)} + \frac{p_2 s_1^{\ 4}\eta_P(\rho+\omega+p_2)^2((\rho+\omega)g_S + p_2 g_p)}{\omega\delta(\rho+\delta)^3(\rho+\omega)^3\eta_K\eta_P\eta_S^{\ 2}} + \frac{p_2 s_1^{\ 2}(\rho+\omega+p_2)\eta_S g_p p_1^{\ 2}(\rho+\delta)^2}{\omega\delta(\rho+\delta)^3(\rho+\omega)^3\eta_K\eta_P\eta_S^{\ 2}} \end{aligned} \quad \text{EqA23}$$

Appendix B Proof of proposition 2

$$\max.\ R_M = \int_0^{+\infty} e^{-\rho t}\left[\pi_M D(t) - \frac{1}{2}\eta_S E_S^2(t) - \frac{1}{2}\eta_P E_P^2(t) + k(t)(S(t)-\overline{S}) + k(t)(P(t)-\overline{P})\right]dt$$

$$\max.\ R_R = \int_0^{+\infty} e^{-\rho t}\left[\pi_R D(t) - \frac{1}{2}\eta_R E_R^2(t)\right]dt$$

The optimal value function of the retailer at t-moment can be presented as:

$$\mathrm{V}_R(S,P) = \max_{E_R(t)} \int_t^{+\infty} e^{-\rho(\tau-t)}\left\{\pi_R D(t) - \frac{1}{2}\eta_R E_R^2(t)\right\}dt \qquad \text{EqB1}$$

For any S⩾0, P⩾0, $\mathrm{V}_R(S,P)$ satisfies the HJB condition as follows:

$$\rho \mathrm{V}_R(S,P) = \max_{E_P(t),E_R(t)}\left\{\pi_R(D_0 + d_1 S(t) + d_2 P(t) + d_3 E_R(t)) - \frac{1}{2}\eta_R E_R^2(t) + V_{RS}^{'}\left[s_1 E_S(t) - \delta S(t)\right] + V_{RP}^{'}\left[p_1 E_P(t) + p_2 S(t) - \omega P(t)\right]\right\}dt \qquad \text{EqB2}$$

The optimal value strategy satisfies that $\frac{\partial^2 \rho V_R(X,S)}{\partial {E_R}^2} = -\eta_R < 0$. For that $\frac{\partial \rho V_R(X,S)}{\partial E_R} = \pi_M d_3 - \eta_R E_R = 0$, we can obtain $E_R(t) = \frac{\pi_R d_3}{\eta_R}$.

Then, the manufacturer make decision on $E_S(t)$ and $E_P(t)$ to maximize its own profit, and the optimal value function of manufacturer at t-moment can be presented as:

$$\mathrm{V}_M(S,P) = \max_{E_S(t),E_P(t)} \int_t^{+\infty} e^{-\rho(\tau-t)}\left\{\pi_M D(t) - \frac{1}{2}\eta_S E_S^2(t) - \frac{1}{2}\eta_P E_P^2(t) + k(t)(S(t)-\overline{S}) + k(t)(P(t)-\overline{P})\right\}dt \qquad \text{EqB3}$$

For any S⩾0, P⩾0, $\mathrm{V}_M(S,P)$ satisfies the HJB condition as follows:

$$\rho \mathrm{V}_M(S,P) = \max_{E_S(t)} \left\{ \begin{array}{l} \pi_M(D_0 + d_1 S(t) + d_2 P(t) + d_3 E_R(t)) - \frac{1}{2}\eta_S E_S^2(t) - \frac{1}{2}\eta_P E_P^2(t) + k(t)(S(t) - \overline{S}) + k(t)(P(t) - \overline{P}) + V_{MS}^{'}\left[s_1 E_S(t) - \delta S(t)\right] \\ + V_{MP}^{'}\left[p_1 E_P(t) + p_2 S(t) - \omega P(t)\right] \end{array} \right\} dt \qquad \text{EqB4}$$

The optimal value strategy satisfies that $\frac{\partial^2 \rho V_M(X,S)}{\partial {E_S}^2} = -\eta_S < 0$ and $\frac{\partial^2 \rho V_M(X,S)}{\partial {E_P}^2} = -\eta_P < 0$ Intuitively, the second-order derivative is negative, which indicates that $\rho \mathrm{V}_M(S,P)$ is concave on $E_S(t), E_P(t)$. The Hessian matrix of $E_S(t), E_P(t)$ is H=$\begin{bmatrix} -\eta_S & 0 \\ 0 & -\eta_P \end{bmatrix}$ >0, due to $-\eta_S$<0, the optimal strategy for the global maximum can be calculated by solving the following equation:

$$\begin{cases} \frac{\partial \rho V_M(X,S)}{\partial E_P} = -\eta_P E_P(t) + p_1 V_{MP}^{'} = 0 \\ \frac{\partial \rho V_M(X,S)}{\partial E_S} = -\eta_S E_S(t) + s_1 V_{MS}^{'} = 0 \end{cases} \qquad \text{EqB5}$$

therefore, $E_S(t) = \frac{s_1 V_{MS}^{'}}{\eta_S}$ and $E_P(t) = \frac{p_1 V_{MP}^{'}}{\eta_P}$.

Substitute $E_S(t) = \frac{s_1 V_{MS}^{'}}{\eta_S}, E_P(t) = \frac{p_1 V_{MP}^{'}}{\eta_P}$ and $E_R(t) = \frac{\pi_R d_3}{\eta_R}$ into EqB4 and simplify as below

$$\rho \mathrm{V}_M(S,P) = \left(\pi_M d_1 + k(t) - \delta V_{MS}^{'} + p_2 V_{MP}^{'}\right) S(t) + \left(\pi_M d_2 - \omega V_{MP}^{'} + k(t)\right) P(t) + \frac{\pi_R \pi_M {d_3}^2}{\eta_R} + \pi_M D_0 + \frac{{s_1}^2 {V_{MS}^{'}}^2}{2\eta_S} + \frac{{p_1}^2 {V_{MP}^{'}}^2}{2\eta_P} - k(t)\overline{S} - k(t)\overline{P} \qquad \text{EqB6}$$

Since $\mathrm{V}_M(S,P)$ is the linear function of $S(t)$ and $P(t)$, we assume that $V_M(S,P) = a_1 S(t) + b_1 P(t) + c_1$, then $V_{MS}^{'} = a_1, V_{MP}^{'} = b_1$, where a_1, b_1, c_1 are constants. Substitute $V_M(S,P)$ and its first partial derivative into EqB6 and simplify, the following equation can be obtained:

$$\begin{cases} \rho a_1 = \pi_M d_1 + k(t) - \delta a_1 + p_2 b_1 \\ \rho b_1 = \pi_M d_2 - \omega b_1 + k(t) \\ \rho c_1 = \dfrac{\pi_R \pi_M {d_3}^2}{\eta_R} + \pi_M D_0 + \dfrac{{s_1}^2 {a_1}^2}{2\eta_S} + \dfrac{{p_1}^2 {b_1}^2}{2\eta_P} - k(t)\overline{S} - k(t)\overline{P} \end{cases} \quad \text{EqB7}$$

the parameter values of a_1, b_1, c_1 can be solved as:

$$\begin{cases} a_1 = \dfrac{(\rho+\omega)\left(\pi_M d_1 + k(t)\right) + p_2\left(\pi_M d_2 + k(t)\right)}{(\rho+\delta)(\rho+\omega)} \\ b_1 = \dfrac{\pi_M d_2 + k(t)}{\rho+\omega} \\ c_1 = \dfrac{\pi_R \pi_M {d_3}^2}{\rho \eta_R} + \dfrac{\pi_M D_0}{\rho} + \dfrac{{s_1}^2\left((\rho+\omega)\left(\pi_M d_1 + k(t)\right) + p_2\left(\pi_M d_2 + k(t)\right)\right)^2}{2\rho\eta_S(\rho+\delta)^2(\rho+\omega)^2} + \dfrac{{p_1}^2\left(\pi_M d_2 + k(t)\right)^2}{2\rho\eta_P(\rho+\omega)^2} - \dfrac{k(t)\overline{S}}{\rho} - \dfrac{k(t)\overline{P}}{\rho} \end{cases} \quad \text{EqB8}$$

Therefore, $E_S(t) = \dfrac{s_1\left((\rho+\omega)\left(\pi_M d_1 + k(t)\right) + p_2\left(\pi_M d_2 + k(t)\right)\right)}{\eta_S(\rho+\delta)(\rho+\omega)}$.

Substitute $E_S(t) = \dfrac{s_1\left((\rho+\omega)\left(\pi_M d_1 + k(t)\right) + p_2\left(\pi_M d_2 + k(t)\right)\right)}{\eta_S(\rho+\delta)(\rho+\omega)}$, $E_P(t) = \dfrac{p_1\left(\pi_M d_2 + k(t)\right)}{\eta_P\left(\rho+\omega\right)}$ and $E_R(t) = \dfrac{\pi_R d_3}{\eta_R}$ into EqB2 and simplify:

$$\rho \mathrm{V}_R(S,P)=\left(\pi_R d_1-\delta V_{RS}^{'}+p_2 V_{RP}^{'}\right)S(t)+\left(\pi_R d_2-\omega V_{RP}^{'}\right)P(t)+\pi_R D_0+\frac{\pi_R^{\ 2} d_3^{\ 2}}{2\eta_R}+V_{RS}^{'}\frac{s_1^{\ 2}\left((\rho+\omega)\left(\pi_M d_1+k(t)\right)+p_2\left(\pi_M d_2+k(t)\right)\right)}{\eta_S(\rho+\delta)(\rho+\omega)}$$

$$+V_{RP}^{'}\frac{p_1^{\ 2}\left(\pi_M d_2+k(t)\right)}{\eta_P\left(\rho+\omega\right)} \qquad \text{EqB9}$$

Since $\mathrm{V}_R(S,P)$ is the linear function of $S(t)$ and $P(t)$, we assume that $\mathrm{V}_R(S,P)=a_2 S(t)+b_2 P(t)+c_2$, then $V_{RS}^{'}=a_2, V_{RP}^{'}=b_2$, where a_2,b_2,c_2 are constants. Substitute $V_R(S,P)$ and its first partial derivative into EqB9 and simplify, the following equation can be obtained:

$$\begin{cases} \rho a_2=\pi_R d_1-\delta a_2+p_2 b_2 \\ \rho b_2=\pi_R d_2-\omega b_2 \\ \rho c_2=\pi_R D_0+\dfrac{\pi_R^{\ 2} d_3^{\ 2}}{2\eta_R}+V_{RS}^{'}\dfrac{s_1^{\ 2}\left((\rho+\omega)\left(\pi_M d_1+k(t)\right)+p_2\left(\pi_M d_2+k(t)\right)\right)}{\eta_S(\rho+\delta)(\rho+\omega)}+V_{RP}^{'}\dfrac{p_1^{\ 2}\left(\pi_M d_2+k(t)\right)}{\eta_P\left(\rho+\omega\right)} \end{cases} \qquad \text{EqB10}$$

the parameter values of a_2,b_2,c_2 can be solved as:

$$\begin{cases} a_2=\dfrac{(\rho+\omega)\pi_R d_1+p_2\pi_R d_2}{(\rho+\delta)(\rho+\omega)} \\ b_2=\dfrac{\pi_R d_2}{\rho+\omega} \\ c_2=\dfrac{\pi_R D_0}{\rho}+\dfrac{\pi_R^{\ 2} d_3^{\ 2}}{2\eta_R\rho}+\dfrac{\left((\rho+\omega)\pi_R d_1+p_2\pi_R d_2\right)s_1^{\ 2}\left((\rho+\omega)\left(\pi_M d_1+k(t)\right)+p_2\left(\pi_M d_2+k(t)\right)\right)}{\rho\eta_S(\rho+\delta)^2(\rho+\omega)^2}+\dfrac{\pi_R d_2 p_1^{\ 2}\left(\pi_M d_2+k(t)\right)}{\rho\eta_P\left(\rho+\omega\right)^2} \end{cases} \qquad \text{EqB11}$$

Therefore,

$$\begin{cases} E_S(t)=\dfrac{s_1\left((\rho+\omega)\left(\pi_M d_1+k(t)\right)+p_2\left(\pi_M d_2+k(t)\right)\right)}{\eta_S(\rho+\delta)(\rho+\omega)} \\ E_P(t)=\dfrac{p_1\left(\pi_M d_2+k(t)\right)}{\eta_P\left(\rho+\omega\right)} \\ E_R(t)=\dfrac{\pi_R d_3}{\eta_R} \end{cases} \qquad \text{EqB12}$$

Then, the government make decision on optimal policy strategy $k(t)$ to maximize its own profit, and the optimal value function of government at t-moment can be presented as:

$$V_G(S,P)=\max_{k(t)}\int_0^{+\infty}e^{-\rho(\tau-t)}\left\{g_s S(t)+g_p P(t)-\frac{1}{2}\eta_k k^2(\mathrm{t})\right\}dt \qquad \text{EqB13}$$

For any S⩾0, P⩾0, $\mathrm{V}_G(S,P)$ satisfies the HJB condition as follows:

$$\rho\mathrm{V}_G(S,P)=\max_{k(t)}\left\{g_s S(t)+g_p P(t)-\frac{1}{2}\eta_k k^2(\mathrm{t})+V_{GS}^{'}\left[s_1 E_S(t)-\delta S(t)\right]+V_{GP}^{'}\left[p_1 E_P(t)+p_2 S(t)-\omega P(t)\right]\right\}dt \qquad \text{EqB14}$$

Substitute $E_S(t),E_P(t)$ and $E_R(t)$ into EqB14, we can obtain:

$$\rho V_G(S,P)=\max_{k(t)}\int_t^{+\infty}e^{-\rho(\tau-t)}\left\{\begin{aligned}&g_s S(t)+g_p P(t)-\frac{1}{2}\eta_k k^2(\mathrm{t})+V_{GS}^{'}\left[\frac{s_1^{\,2}\left((\rho+\omega)\left(\pi_M d_1+k(t)\right)+p_2\left(\pi_M d_2+k(t)\right)\right)}{\eta_S(\rho+\delta)(\rho+\omega)}-\delta S(t)\right]\\&+V_{GP}^{'}\left[\frac{p_1^{\,2}\left(\pi_M d_2+k(t)\right)}{\eta_P(\rho+\omega)}+p_2 S(t)-\omega P(t)\right]\end{aligned}\right\}dt \qquad \text{EqB15}$$

Simplify EqB15 to a function with respect to government strategy k(t) can obtain:

$$\rho V_G(S,P)=\max_{k(t)}\int_t^{+\infty}e^{-\rho(\tau-t)}\left\{\begin{aligned}&\left(-\frac{1}{2}\eta_k\right)k^2(t)+\left(\frac{V'_{GS}s_1^{\,2}(\rho+\omega+p_2)}{\eta_S(\rho+\delta)(\rho+\omega)}+\frac{V'_{GP}p_1^{\,2}}{\eta_P(\rho+\omega)}\right)k(t)+g_sS(t)+g_pP(t)+\frac{V'_{GS}s_1^{\,2}\left((\rho+\omega)\pi_Md_1+p_2\pi_Md_2\right)}{\eta_S(\rho+\delta)(\rho+\omega)}-V'_{GS}\delta S(t)\\&+\frac{V'_{GP}p_1^{\,2}\pi_Md_2}{\eta_P(\rho+\omega)}+V'_{GP}p_2S(t)-V'_{GP}\omega P(t)\end{aligned}\right\}dt \quad \text{EqB16}$$

Intuitively, the second-order derivative is negative, which indicates that $\rho V_G(S,P)$ is concave on $k(t)$. The necessary condition of optimal value strategy requires $\frac{\partial\rho V_G(S,P)}{\partial k(t)}=0$. By solving this equation, the optimal strategy $k^*(t)$ for the global maximum can be calculated.

$$k^*(t)=\frac{V'_{GS}s_1^{\,2}(\rho+\omega+p_2)}{\eta_k\eta_S(\rho+\delta)(\rho+\omega)}+\frac{V'_{GP}p_1^{\,2}}{\eta_k\eta_P(\rho+\omega)}=\frac{V'_{GS}s_1^{\,2}\eta_P(\rho+\omega+p_2)+V'_{GP}p_1^{\,2}\eta_S(\rho+\delta)}{\eta_k\eta_P\eta_S(\rho+\delta)(\rho+\omega)} \quad \text{EqB17}$$

Since $V_G(S,P)$ is the linear function of $S(t)$ and $P(t)$, we assume that $V_G(S,P)=AS(t)+BP(t)+C$, then $V'_{GS}=A, V'_{GP}=B$, where A,B,C are constants. Substitute $V_G(S,P)$ and its first partial derivative into EqB16 and simplify, the following equation can be obtained:

$$\rho\left[AS(t)+BP(t)+C\right]=\left(g_s-\delta V'_{GS}+p_2V'_{GP}\right)S(t)+\left(g_p-\omega V'_{GP}\right)P(t)-\frac{1}{2}\eta_kk^2(\text{t})+\frac{V'_{GS}s_1^{\,2}\left((\rho+\omega)\left(\pi_Md_1+k(t)\right)+p_2\left(\pi_Md_2+k(t)\right)\right)}{\eta_S(\rho+\delta)(\rho+\omega)}+\frac{V'_{GP}p_1^{\,2}\left(\pi_Md_2+k(t)\right)}{\eta_P(\rho+\omega)} \quad \text{EqB18}$$

Using the method of undetermined coefficients, we can get the following ternary linear functions:

$$\begin{cases}\rho A=g_s-\delta A+p_2B\\ \rho B=g_p-\omega B\\ \rho C=\dfrac{As_1^{\,2}\left((\rho+\omega)\left(\pi_Md_1+k(t)\right)+p_2\left(\pi_Md_2+k(t)\right)\right)}{\eta_S(\rho+\delta)(\rho+\omega)}+\dfrac{Bp_1^{\,2}\left(\pi_Md_2+k(t)\right)}{\eta_P(\rho+\omega)}-\dfrac{1}{2}\eta_kk^2(\text{t})\end{cases} \quad \text{EqB19}$$

the parameter values of A,B,C can be solved as:

$$\begin{cases} A = \dfrac{(\rho+\omega)g_S + p_2 g_p}{(\rho+\delta)(\rho+\omega)} \\ B = \dfrac{g_p}{\rho+\omega} \\ C = \dfrac{\left((\rho+\omega)g_S + p_2 g_p\right)s_1^{\ 2}\left((\rho+\omega)\left(\pi_M d_1 + k(t)\right) + p_2\left(\pi_M d_2 + k(t)\right)\right)}{\rho\eta_S(\rho+\delta)^2(\rho+\omega)^2} + \dfrac{g_p p_1^{\ 2}\left(\pi_M d_2 + k(t)\right)}{\rho\eta_P(\rho+\omega)^2} - \dfrac{\eta_k k^2(\mathrm{t})}{2\rho} \end{cases} \qquad \text{EqB20}$$

The optimal strategy of the government, retailer and the manufacturer are as below:

$$k^*(t) = \frac{s_1^{\ 2}(\rho+\omega+p_2)\left((\rho+\omega)g_S + p_2 g_p\right)}{\eta_K\eta_S(\rho+\delta)^2(\rho+\omega)^2} + \frac{g_p p_1^{\ 2}}{\eta_K\eta_P(\rho+\omega)^2} = \frac{\eta_P s_1^{\ 2}(\rho+\omega+p_2)\left((\rho+\omega)g_S + p_2 g_p\right) + \eta_S g_p p_1^{\ 2}(\rho+\delta)^2}{\eta_P\eta_K\eta_S(\rho+\delta)^2(\rho+\omega)^2}$$

$$\begin{cases} E_S(t) = \dfrac{s_1\left(\pi_M d_1(\rho+\omega) + \pi_M d_2 p_2\right)}{\eta_S(\rho+\delta)(\rho+\omega)} + \dfrac{s_1\left(\rho+\omega+p_2\right)\left(\eta_P s_1^{\ 2}(\rho+\omega+p_2)\left((\rho+\omega)g_S + p_2 g_p\right) + \eta_S g_p p_1^{\ 2}(\rho+\delta)^2\right)}{\eta_S^{\ 2}\eta_P\eta_K(\rho+\delta)^3(\rho+\omega)^3} \\ E_P(t) = \dfrac{p_1\pi_M d_2}{\eta_P\left(\rho+\omega\right)} + \dfrac{p_1\eta_P s_1^{\ 2}(\rho+\omega+p_2)\left((\rho+\omega)g_S + p_2 g_p\right) + \eta_S g_p p_1^{\ 3}(\rho+\delta)^2}{\eta_P^{\ 2}\eta_K\eta_S(\rho+\delta)^2(\rho+\omega)^3} \\ E_R(t) = \dfrac{\pi_R d_3}{\eta_R} \end{cases} \qquad \text{EqB21}$$

Substitute $E_S(t), E_P(t)$ and $E_R(t)$ into $\dot{S}(t)$ and $\dot{P}(t)$, we can obtain:

$$\begin{cases} \dot{S}(t) = \dfrac{s_1^{\ 2}\left(\pi_M d_1(\rho+\omega) + \pi_M d_2 p_2\right)}{\eta_S(\rho+\delta)(\rho+\omega)} + \dfrac{s_1^{\ 2}\left(\rho+\omega+p_2\right)\left(\eta_P s_1^{\ 2}(\rho+\omega+p_2)\left((\rho+\omega)g_S + p_2 g_p\right) + \eta_S g_p p_1^{\ 2}(\rho+\delta)^2\right)}{\eta_S^{\ 2}\eta_P\eta_K(\rho+\delta)^3(\rho+\omega)^3} - \delta S(t) \\ \dot{P}(t) = \dfrac{p_1^{\ 2}\pi_M d_2}{\eta_P\left(\rho+\omega\right)} + \dfrac{p_1^{\ 2}\eta_P s_1^{\ 2}(\rho+\omega+p_2)\left((\rho+\omega)g_S + p_2 g_p\right) + \eta_S g_p p_1^{\ 4}(\rho+\delta)^2}{\eta_P^{\ 2}\eta_K\eta_S(\rho+\delta)^2(\rho+\omega)^3} + p_2 S(t) - \omega P(t) \end{cases} \qquad \text{EqB22}$$

The state of trajectory of the digital level can be obtained by solving the first-order differential equation EqB21, and the it can be presented as:

$S^*(t) = S_\infty + (S_0 - S_\infty)e^{-\delta t}$, where

$$S_{\infty}=\frac{s_1^{\ 2}\left(\pi_M d_1(\rho+\omega)+\pi_M d_2 p_2\right)}{\delta\eta_S(\rho+\delta)(\rho+\omega)}+\frac{s_1^{\ 2}\left(\rho+\omega+p_2\right)\left(\eta_P s_1^{\ 2}(\rho+\omega+p_2)\left((\rho+\omega)g_S+p_2 g_p\right)+\eta_S g_p p_1^{\ 2}(\rho+\delta)^2\right)}{\delta\eta_S^{\ 2}\eta_P\eta_K(\rho+\delta)^3(\rho+\omega)^3} \quad \text{EqB23}$$

Substitute $S^*(t)$ into $\dot{P}(t)$, we can obtain that

$$\dot{P}(t)=\frac{p_1^{\ 2}\pi_M d_2}{\eta_P\left(\rho+\omega\right)}+\frac{p_1^{\ 2}\eta_P s_1^{\ 2}(\rho+\omega+p_2)\left((\rho+\omega)g_S+p_2 g_p\right)+\eta_S g_p p_1^{\ 4}(\rho+\delta)^2}{\eta_P^{\ 2}\eta_K\eta_S(\rho+\delta)^2(\rho+\omega)^3}+p_2(S_{\infty}+(S_0-S_{\infty})e^{-\delta t})-\omega P(t) \quad \text{EqB24}$$

The state of trajectory of the carbon adatement level can be obtained by solving the first-order differential equation EqB23, and the it can be presented as:

$P^*(t)=P_{\infty}+\left(P_0-P_{\infty}-\dfrac{p_2(S_0-S_{\infty})}{\omega-\delta}\right)e^{-\omega t}+\dfrac{p_2(S_0-S_{\infty})}{\omega-\delta}e^{-\delta t}$, where

$$\begin{aligned}P_{\infty}=&\frac{p_1^{\ 2}\pi_M d_2}{\omega\eta_P\left(\rho+\omega\right)}+\frac{p_1^{\ 2}\eta_P s_1^{\ 2}(\rho+\omega+p_2)\left((\rho+\omega)g_S+p_2 g_p\right)+\eta_S g_p p_1^{\ 4}(\rho+\delta)^2}{\omega\eta_P^{\ 2}\eta_K\eta_S(\rho+\delta)^2(\rho+\omega)^3}+\frac{p_2 s_1^{\ 2}\left(\pi_M d_1(\rho+\omega)+\pi_M d_2 p_2\right)}{\omega\delta\eta_S(\rho+\delta)(\rho+\omega)}\\&+\frac{p_2 s_1^{\ 2}\left(\rho+\omega+p_2\right)\left(\eta_P s_1^{\ 2}(\rho+\omega+p_2)\left((\rho+\omega)g_S+p_2 g_p\right)+\eta_S g_p p_1^{\ 2}(\rho+\delta)^2\right)}{\omega\delta\eta_S^{\ 2}\eta_P\eta_K(\rho+\delta)^3(\rho+\omega)^3}\end{aligned} \quad \text{EqB25}$$

Appendix C Proof of proposition 3

$$\max.\ R_M=\int_0^{+\infty}e^{-\rho t}\left[\pi_M D(t)-\frac{1}{2}\eta_S E_S^2(t)-\frac{1}{2}\eta_P E_P^2(t)-\frac{1}{2}\psi_R\eta_R E_R^2(t)+k(t)(S(t)-\overline{S})+k(t)(P(t)-\overline{P})\right]dt$$

$$\max.\ R_R=\int_0^{+\infty}e^{-\rho t}\left[\pi_R D(t)-\frac{1}{2}(1-\psi_R)\eta_R E_R^2(t)\right]dt$$

The optimal value function of the retailer at t-moment can be presented as:

$$\mathrm{V}_R(S,P)=\max_{E_R}\int_t^{+\infty}e^{-\rho(\tau-t)}\left\{\pi_R D(t)-\frac{1}{2}(1-\psi_R)\eta_R E_R^2(t)\right\}dt \quad \text{EqC1}$$

For any S⩾0, P⩾0, $\mathrm{V}_R(S,P)$ satisfies the HJB condition as follows:

$$\rho \mathrm{V}_R(S,P)=\max_{E_R(t)}\left\{\pi_R(D_0+d_1S(t)+d_2P(t)+d_3E_R(t))-\frac{1}{2}(1-\psi_R)\eta_R E_R^2(t)+V_{RS}^{'}\left[s_1E_S(t)-\delta S(t)\right]+V_{RP}^{'}\left[p_1E_P(t)+p_2S(t)-\omega P(t)\right]\right\}dt \qquad \text{EqC2}$$

The optimal value strategy satisfies that $\frac{\partial^2 \rho V_R(X,S)}{\partial {E_R}^2}=-(1-\psi_R)\eta_R<0$, which indicates that $\rho \mathrm{V}_R(S,P)$ is concave on E_R . For that, $\frac{\partial \rho V_R(X,S)}{\partial E_R}=-(1-\psi_R)\eta_R E_R+\pi_R d_3=0$, we have $E_R(t)=\frac{\pi_R d_3}{(1-\psi_R)\eta_R}$.

Then, the manufacturer make decision on $E_S(t)$, $E_P(t)$ and ψ_R to maximize its own profit, and the optimal value function of manufacturer at t-moment can be presented as:

$$\mathrm{V}_M(S,P)=\max_{E_S(t)}\int_t^{+\infty}e^{-\rho(\tau-t)}\left\{\pi_M D(t)-\frac{1}{2}\eta_S E_S^2(t)-\frac{1}{2}\eta_P E_P^2(t)-\frac{1}{2}\psi_R\eta_R E_R^2(t)+k(t)(S(t)-\overline{S})+k(t)(P(t)-\overline{P})\right\}dt \qquad \text{EqC3}$$

For any S⩾0, P⩾0, $\mathrm{V}_M(S,P)$ satisfies the HJB condition as follows:

$$\rho \mathrm{V}_M(S,P)=\max_{E_S(t)}\left\{\begin{aligned}&\pi_M(D_0+d_1S(t)+d_2P(t)+\frac{\pi_R {d_3}^2}{(1-\psi_R)\eta_R})-\frac{1}{2}\eta_S E_S^2(t)-\frac{1}{2}\eta_P E_P^2(t)-\frac{\psi_R {\pi_R}^2 {d_3}^2}{2(1-\psi_R)^2\eta_R}+k(t)(S(t)-\overline{S})+k(t)(P(t)-\overline{P})\\&+V_{MS}^{'}\left[s_1E_S(t)-\delta S(t)\right]+V_{MP}^{'}\left[p_1E_P(t)+p_2S(t)-\omega P(t)\right]\end{aligned}\right\}dt \qquad \text{EqC4}$$

The optimal value strategy satisfies $\frac{\partial^2 \rho V_M(X,S)}{\partial {E_P}^2}=-\eta_P<0$, $\frac{\partial^2 \rho V_M(X,S)}{\partial {E_S}^2}=-\eta_S<0$ and

$$\frac{\partial^2 \rho V_M(X,S)}{\partial {\psi_R}^2}=\frac{2\pi_M\pi_R {d_3}^2}{(1-\psi_R)^3\eta_R}-\frac{{\pi_R}^2{d_3}^2}{2\eta_R}\left(\frac{4(1-\psi_R)+6\psi_R}{(1-\psi_R)^4}\right)=-\frac{{\pi_R}^2{d_3}^2}{\eta_R(1-\psi_R)^4}<0$$

which indicates $\rho \mathrm{V}_S(S,P)$ is concave on $E_P(t), E_S(t)$ and ψ_R. The Hessian matrix of $E_P(t), E_S(t)$ and ψ_R is $H=\begin{bmatrix} -\eta_P & & \\ & -\eta_S & \\ & & -\dfrac{\pi_R{}^2 d_3{}^2}{\eta_R(1-\psi_R)^4} \end{bmatrix} > 0$,

and $-\eta_P < 0$, the $E_S(t),\ E_P(t)$ and ψ_R can be calculated by solving the following equations:

$$\begin{cases} \dfrac{\partial \rho V_S(X,S)}{\partial E_S} = -\eta_S E_S(t) + s_1 V_{MS}^{'} = 0 \\ \dfrac{\partial \rho V_S(X,S)}{\partial E_P} = -\eta_P E_P(t) + p_1 V_{MP}^{'} = 0 \\ \dfrac{\partial \rho V_S(X,S)}{\partial \psi_R} = \dfrac{\pi_M \pi_R d_3{}^2}{(1-\psi_R)^2 \eta_R} - \dfrac{\pi_R{}^2 d_3{}^2}{2\eta_R}\left(\dfrac{1}{(1-\psi_R)^2} + \dfrac{2\psi_R}{(1-\psi_R)^3}\right) = 0 \end{cases} \qquad \text{EqC5}$$

Then, we can obtain:

$$\begin{cases} E_S(t) = \dfrac{s_1 V_{MS}^{'}}{\eta_S} \\ E_P(t) = \dfrac{p_1 V_{MP}^{'}}{\eta_P} \\ \psi_R = \dfrac{2\pi_M - \pi_R}{2\pi_M + \pi_R}\left(2\pi_M > \pi_R, \psi_M > 0\right) \end{cases} \qquad \text{EqC6.}$$

Substitute $E_S, E_P(t)$, $E_R(t)$ and ψ_R into EqC4 and simplify:

$$\rho V_M(S,P)=\left(\pi_M d_1+k(t)-\delta V_{MS}^{'}+p_2V_{MP}^{'}\right)S(t)+\left(\pi_M d_2+k(t)-\omega V_{MP}^{'}\right)P(t)+\frac{\left(2\pi_M+\pi_R\right)^2 d_3^{\ 2}}{8\eta_R}+\pi_M D_0+\frac{s_1^{\ 2}V_{MS}^{'\ 2}}{2\eta_S}+\frac{p_1^{\ 2}V_{MP}^{'\ 2}}{2\eta_P}-k(t)\overline{S}-k(t)\overline{P} \qquad \text{EqC7}$$

Since $V_M(S,P)$ is the linear function of $S(t)$ and $P(t)$, we assume that $V_M(S,P)=a_1S(t)+b_1P(t)+c_1$, then $V_{MS}^{'}=a_1,V_{MP}^{'}=b_1$, where a_1,b_1,c_1 are constants. Substitute $V_M(S,P)$ and its first partial derivative into EqC8 and simplify, the following equation can be obtained:

$$\begin{cases}\rho a_1=\pi_M d_1+k(t)-\delta a_1+p_2b_1\\ \rho b_1=\pi_M d_2+k(t)-\omega b_1\\ \rho c_1=\dfrac{\left(2\pi_M+\pi_R\right)^2 d_3^{\ 2}}{8\eta_R}+\pi_M D_0+\dfrac{s_1^{\ 2}a_1^{\ 2}}{2\eta_S}+\dfrac{p_1^{\ 2}b_1^{\ 2}}{2\eta_P}-k(t)\overline{S}-k(t)\overline{P}\end{cases} \qquad \text{EqC8}$$

the parameter values of a_1,b_1,c_1 can be solved as:

$$\begin{cases}a_1=\dfrac{(\rho+\omega)\left(\pi_M d_1+k(t)\right)+p_2\left(\pi_M d_2+k(t)\right)}{(\rho+\delta)(\rho+\omega)}\\ b_1=\dfrac{\pi_M d_2+k(t)}{\rho+\omega}\\ c_1=\dfrac{\left(2\pi_M+\pi_R\right)^2 d_3^{\ 2}}{8\rho\eta_R}+\dfrac{\pi_M D_0}{\rho}+\dfrac{s_1^{\ 2}\left((\rho+\omega)\left(\pi_M d_1+k(t)\right)+p_2\left(\pi_M d_2+k(t)\right)\right)^2}{2\rho\eta_S(\rho+\delta)^2(\rho+\omega)^2}+\dfrac{p_1^{\ 2}\left(\pi_M d_2+k(t)\right)^2}{2\rho\eta_P\left(\rho+\omega\right)^2}-\dfrac{k(t)\overline{S}}{\rho}-\dfrac{k(t)\overline{P}}{\rho}\end{cases} \qquad \text{EqC9}$$

Therefore,

$$\begin{cases} E_S(t) = \dfrac{s_1\left((\rho+\omega)\left(\pi_M d_1 + k(t)\right) + p_2\left(\pi_M d_2 + k(t)\right)\right)}{\eta_S(\rho+\delta)(\rho+\omega)} \\ E_P(t) = \dfrac{p_1\left(\pi_M d_2 + k(t)\right)}{\eta_P\left(\rho+\omega\right)} \\ E_R(t) = \dfrac{d_3\left(2\pi_M + \pi_R\right)}{2\eta_R} \end{cases} \qquad \text{EqC10}$$

Substitute $E_S, E_P(t)$, $E_R(t)$ and ψ_M into EqC2 and simplify:

$$\rho \mathrm{V}_R(S,P) = \left(\pi_R d_1 - \delta V_{RS}^{'} + p_2 V_{RP}^{'}\right)S(t) + \left(\pi_R d_2 - \omega V_{RP}^{'}\right)P(t) + \pi_R D_0 + \frac{\pi_R d_3^{\ 2}\left(2\pi_M + \pi_R\right)}{4\eta_R} + \frac{V_{RS}^{'} s_1^{\ 2}\left((\rho+\omega)\left(\pi_M d_1 + k(t)\right) + p_2\left(\pi_M d_2 + k(t)\right)\right)}{\eta_S(\rho+\delta)(\rho+\omega)} + \frac{V_{RP}^{'} p_1^{\ 2}\left(\pi_M d_2 + k(t)\right)}{\eta_P\left(\rho+\omega\right)}$$

EqC11

Since $\mathrm{V}_R(S,P)$ is the linear function of $S(t)$ and $P(t)$, we assume that $\mathrm{V}_R(S,P) = a_2 S(t) + b_2 P(t) + c_2$, then $V_{RS}^{'} = a_2, V_{RP}^{'} = b_2$, where a_2, b_2, c_2 are constants. Substitute $V_R(S,P)$ and its first partial derivative into EqC11 and simplify, the following equation can be obtained:

$$\begin{cases} \rho a_2 = \pi_R d_1 - \delta a_2 + p_2 b_2 \\ \rho b_2 = \pi_R d_2 - \omega b_2 \\ \rho c_2 = \pi_R D_0 + \dfrac{\pi_R d_3^{\ 2}\left(2\pi_M + \pi_R\right)}{4\eta_R} + \dfrac{V_{RS}^{'} s_1^{\ 2}\left((\rho+\omega)\left(\pi_M d_1 + k(t)\right) + p_2\left(\pi_M d_2 + k(t)\right)\right)}{\eta_S(\rho+\delta)(\rho+\omega)} + \dfrac{V_{RP}^{'} p_1^{\ 2}\left(\pi_M d_2 + k(t)\right)}{\eta_P\left(\rho+\omega\right)} \end{cases} \qquad \text{EqC12}$$

the parameter values of a_2, b_2, c_2 can be solved as:

$$\begin{cases} a_2 = \dfrac{(\rho+\omega)\pi_R d_1 + p_2 \pi_R d_2}{(\rho+\delta)(\rho+\omega)} \\ b_2 = \dfrac{\pi_R d_2}{\rho+\omega} \\ c_2 = \dfrac{\pi_R D_0}{\rho} + \dfrac{\pi_R d_3^{\,2}\left(2\pi_M + \pi_R\right)}{4\rho\eta_R} + \dfrac{s_1^{\,2}\left((\rho+\omega)\pi_R d_1 + p_2\pi_R d_2\right)\left((\rho+\omega)\left(\pi_M d_1 + k(t)\right) + p_2\left(\pi_M d_2 + k(t)\right)\right)}{\rho\eta_S(\rho+\delta)^2(\rho+\omega)^2} + \dfrac{\pi_R d_2 p_1^{\,2}\left(\pi_M d_2 + k(t)\right)}{\rho\eta_P\left(\rho+\omega\right)^2} \end{cases} \quad \text{EqC13}$$

Then, the government make decision on optimal policy strategy $k(t)$ to maximize its own profit, and the optimal value function of government at t-moment can be presented as:

$$V_G(S,P) = \max_{k(t)} \int_0^{+\infty} e^{-\rho(\tau-t)} \left\{ g_s S(t) + g_p P(t) - \frac{1}{2}\eta_k k^2(\mathrm{t}) \right\} dt \quad \text{EqC14}$$

For any S⩾0, P⩾0, $\mathrm{V}_G(S,P)$ satisfies the HJB condition as follows:

$$\rho \mathrm{V}_G(S,P) = \max_{k(t)} \left\{ g_s S(t) + g_p P(t) - \frac{1}{2}\eta_k k^2(\mathrm{t}) + V_{GS}^{'}\left[s_1 E_S(t) - \delta S(t)\right] + V_{GP}^{'}\left[p_1 E_P(t) + p_2 S(t) - \omega P(t)\right] \right\} dt \quad \text{EqC15}$$

Substitute $E_S(t), E_P(t)$ and $E_R(t)$ into EqC15, we can obtain:

$$\rho V_G(S,P) = \max_{k(t)} \int_t^{+\infty} e^{-\rho(\tau-t)} \left\{ \begin{array}{l} g_s S(t) + g_p P(t) - \dfrac{1}{2}\eta_k k^2(\mathrm{t}) + V_{GS}^{'}\left[\dfrac{s_1^{\,2}\left((\rho+\omega)\left(\pi_M d_1 + k(t)\right) + p_2\left(\pi_M d_2 + k(t)\right)\right)}{\eta_S(\rho+\delta)(\rho+\omega)} - \delta S(t) \right] \\ + V_{GP}^{'}\left[\dfrac{p_1^{\,2}\left(\pi_M d_2 + k(t)\right)}{\eta_P\left(\rho+\omega\right)} + p_2 S(t) - \omega P(t) \right] \end{array} \right\} dt \quad \text{EqC16}$$

Simplify EqC16 to a function with respect to government strategy k(t) can obtain:

$$\rho V_G(S,P)=\max_{k(t)}\int_t^{+\infty}e^{-\rho(\tau-t)}\left\{\begin{array}{l}\left(-\frac{1}{2}\eta_k\right)k^2(t)+\left(\frac{V_{GS}^{'}s_1^{\,2}(\rho+\omega+p_2)}{\eta_S(\rho+\delta)(\rho+\omega)}+\frac{V_{GP}^{'}p_1^{\,2}}{\eta_P(\rho+\omega)}\right)k(t)+g_sS(t)+g_pP(t)+\frac{V_{GS}^{'}s_1^{\,2}\left((\rho+\omega)\pi_Md_1+p_2\pi_Md_2\right)}{\eta_S(\rho+\delta)(\rho+\omega)}+\frac{V_{GP}^{'}p_1^{\,2}\pi_Md_2}{\eta_P(\rho+\omega)}-V_{GS}^{'}\delta S(t)\\+V_{GP}^{'}p_2S(t)-V_{GP}^{'}\omega P(t)\end{array}\right\}dt$$

EqC17

Intuitively, the second-order derivative is negative, which indicates that $\rho \mathrm{V}_G(S,P)$ is concave on $k(t)$. The necessary condition of optimal value strategy requires $\frac{\partial\rho V_G(S,P)}{\partial k(t)}=0$. By solving this equation, the optimal strategy $k^*(t)$ for the global maximum can be calculated as

$$k^*(t)=\frac{V_{GS}^{'}s_1^{\,2}(\rho+\omega+p_2)}{\eta_K\eta_S(\rho+\delta)(\rho+\omega)}+\frac{V_{GP}^{'}p_1^{\,2}}{\eta_K\eta_P(\rho+\omega)} \qquad \text{EqC18}$$

Since $V_G(S,P)$ is the linear function of $S(t)$ and $P(t)$, we assume that $V_G(S,P)=AS(t)+BP(t)+C$, then $V_{GS}^{'}=A, V_{GP}^{'}=B$, where A,B,C are constants. Substitute $V_G(S,P)$ and its first partial derivative into EqC19 and simplify, the following equation can be obtained:

$$\rho\left[AS(t)+BP(t)+C\right]=\left(g_s-\delta V_{GS}^{'}+p_2V_{GP}^{'}\right)S(t)+\left(g_p-\omega V_{GP}^{'}\right)P(t)-\frac{1}{2}\eta_k k^2(\mathrm{t})+\frac{V_{GS}^{'}s_1^{\,2}\left((\rho+\omega)\left(\pi_Md_1+k(t)\right)+p_2\left(\pi_Md_2+k(t)\right)\right)}{\eta_S(\rho+\delta)(\rho+\omega)}+\frac{V_{GP}^{'}p_1^{\,2}\left(\pi_Md_2+k(t)\right)}{\eta_P(\rho+\omega)} \qquad \text{EqC19}$$

Using the method of undetermined coefficients, we can get the following ternary linear functions:

$$\begin{cases}\rho A=g_s-\delta A+p_2B\\ \rho B=g_p-\omega B\\ \rho C=-\frac{1}{2}\eta_k k^2(\mathrm{t})+\frac{As_1^{\,2}\left((\rho+\omega)\left(\pi_Md_1+k(t)\right)+p_2\left(\pi_Md_2+k(t)\right)\right)}{\eta_S(\rho+\delta)(\rho+\omega)}+\frac{Bp_1^{\,2}\left(\pi_Md_2+k(t)\right)}{\eta_P(\rho+\omega)}\end{cases} \qquad \text{EqC20}$$

the parameter values of A, B, C can be solved as:

$$\begin{cases} A = \dfrac{(\rho+\omega)g_S + p_2 g_p}{(\rho+\delta)(\rho+\omega)} \\ B = \dfrac{g_p}{\rho+\omega} \\ C = -\dfrac{1}{2}\eta_k k^2(\mathrm{t}) + \dfrac{\left((\rho+\omega)g_S + p_2 g_p\right)s_1^{\,2}\left((\rho+\omega)\left(\pi_M d_1 + k(t)\right) + p_2\left(\pi_M d_2 + k(t)\right)\right)}{\eta_S(\rho+\delta)^2(\rho+\omega)^2} + \dfrac{g_p p_1^{\,2}\left(\pi_M d_2 + k(t)\right)}{\eta_P(\rho+\omega)^2} \end{cases} \qquad \text{EqC21}$$

The optimal strategy of the government, manufacturer and the retailer are as below:

$$k^*(t) = \frac{\left((\rho+\omega)g_S + p_2 g_p\right)s_1^{\,2}(\rho+\omega+p_2)}{\eta_K\eta_S(\rho+\delta)^2(\rho+\omega)^2} + \frac{g_p p_1^{\,2}}{\eta_K\eta_P(\rho+\omega)^2}$$

$$\begin{cases} E_S(t) = \dfrac{s_1\left(\pi_M d_1(\rho+\omega) + p_2\pi_M d_2\right)}{\eta_S(\rho+\delta)(\rho+\omega)} + \dfrac{\left((\rho+\omega)g_S + p_2 g_p\right)s_1^{\,3}(\rho+\omega+p_2)^2}{\eta_K\eta_S^{\,2}(\rho+\delta)^3(\rho+\omega)^3} + \dfrac{g_p p_1^{\,2} s_1\left(\rho+\omega+p_2\right)}{\eta_S(\rho+\delta)\eta_K\eta_P(\rho+\omega)^3} \\ E_P(t) = \dfrac{p_1\pi_M d_2}{\eta_P\left(\rho+\omega\right)} + \dfrac{p_1\left((\rho+\omega)g_S + p_2 g_p\right)s_1^{\,2}(\rho+\omega+p_2)}{\eta_P\eta_K\eta_S(\rho+\delta)^2(\rho+\omega)^3} + \dfrac{g_p p_1^{\,3}}{\eta_K\eta_P^{\,2}(\rho+\omega)^3} \\ E_R(t) = \dfrac{d_3\left(2\pi_M + \pi_R\right)}{2\eta_R} \end{cases} \qquad \text{EqC22}$$

Substitute $E_S(t), E_P(t)$ and $E_R(t)$ into $\dot{S}(t)$ and $\dot{P}(t)$, we can obtain:

$$\begin{cases} \dot{S}(t) = \dfrac{s_1^{\,2}\left(\pi_M d_1(\rho+\omega) + p_2\pi_M d_2\right)}{\eta_S(\rho+\delta)(\rho+\omega)} + \dfrac{\left((\rho+\omega)g_S + p_2 g_p\right)s_1^{\,4}(\rho+\omega+p_2)^2}{\eta_K\eta_S^{\,2}(\rho+\delta)^3(\rho+\omega)^3} + \dfrac{g_p p_1^{\,2} s_1^{\,2}\left(\rho+\omega+p_2\right)}{\eta_S(\rho+\delta)\eta_K\eta_P(\rho+\omega)^3} - \delta S(t) \\ \dot{P}(t) = \dfrac{p_1^{\,2}\pi_M d_2}{\eta_P\left(\rho+\omega\right)} + \dfrac{p_1^{\,2}\left((\rho+\omega)g_S + p_2 g_p\right)s_1^{\,2}(\rho+\omega+p_2)}{\eta_P\eta_K\eta_S(\rho+\delta)^2(\rho+\omega)^3} + \dfrac{g_p p_1^{\,4}}{\eta_K\eta_P^{\,2}(\rho+\omega)^3} + p_2 S(t) - \omega P(t) \end{cases} \qquad \text{EqC23}$$

The state of trajectory of the digital level can be obtained by solving the first-order differential equation EqC21, and the it can be presented as:

$S^*(t)=S_\infty+(S_0-S_\infty)e^{-\delta t}$, where

$$S_\infty=\frac{s_1^2\left(\pi_M d_1(\rho+\omega)+p_2\pi_M d_2\right)}{\delta\eta_S(\rho+\delta)(\rho+\omega)}+\frac{\left((\rho+\omega)g_S+p_2g_p\right)s_1^4(\rho+\omega+p_2)^2}{\delta\eta_K\eta_S^2(\rho+\delta)^3(\rho+\omega)^3}+\frac{g_p p_1^2 s_1^2\left(\rho+\omega+p_2\right)}{\delta\eta_S(\rho+\delta)\eta_K\eta_P(\rho+\omega)^3} \quad \text{EqC24}$$

Substitute $\dot S(t)$ and $\dot P(t)$, we can obtain that

$$\dot P(t)=\frac{p_1^2\pi_M d_2}{\eta_P\left(\rho+\omega\right)}+\frac{p_1^2\left((\rho+\omega)g_S+p_2g_p\right)s_1^2(\rho+\omega+p_2)}{\eta_P\eta_K\eta_S(\rho+\delta)^2(\rho+\omega)^3}+\frac{g_p p_1^4}{\eta_K\eta_P^2(\rho+\omega)^3}+p_2(S_\infty+(S_0-S_\infty)e^{-\delta t})-\omega P(t) \quad \text{EqC25}$$

The state of trajectory of the carbon adatement level can be obtained by solving the first-order differential equation EqC23, and the it can be presented as:

$P^*(t)=P_\infty+\left(P_0-P_\infty-\dfrac{p_2(S_0-S_\infty)}{\omega-\delta}\right)e^{-\omega t}+\dfrac{p_2(S_0-S_\infty)}{\omega-\delta}e^{-\delta t}$, where

$$\begin{aligned}P_\infty=&\frac{p_1^2\pi_M d_2}{\omega\eta_P\left(\rho+\omega\right)}+\frac{p_1^2\left((\rho+\omega)g_S+p_2g_p\right)s_1^2(\rho+\omega+p_2)}{\omega\eta_P\eta_K\eta_S(\rho+\delta)^2(\rho+\omega)^3}+\frac{g_p p_1^4}{\omega\eta_K\eta_P^2(\rho+\omega)^3}\\&+\frac{p_2s_1^2\left(\pi_M d_1(\rho+\omega)+p_2\pi_M d_2\right)}{\omega\delta\eta_S(\rho+\delta)(\rho+\omega)}+\frac{p_2\left((\rho+\omega)g_S+p_2g_p\right)s_1^4(\rho+\omega+p_2)^2}{\omega\delta\eta_K\eta_S^2(\rho+\delta)^3(\rho+\omega)^3}+\frac{p_2g_p p_1^2 s_1^2\left(\rho+\omega+p_2\right)}{\omega\delta\eta_S(\rho+\delta)\eta_K\eta_P(\rho+\omega)^3}\end{aligned} \quad \text{EqC26}$$

Appendix D Proof of proposition 4

$$\max.\, R_M=\int_0^{+\infty}e^{-\rho t}\left[\pi_M D(t)-\left(1-\phi_{MS}\right)\frac{1}{2}\eta_S E_S^2(t)-\left(1-\phi_{MP}\right)\frac{1}{2}\eta_P E_P^2(t)-\psi_R\frac{1}{2}\eta_R E_R^2(t)+k(t)(S(t)-\overline{S})+k(t)(P(t)-\overline{P})\right]dt$$

$$\max.\, R_R=\int_0^{+\infty}e^{-\rho t}\left[\pi_R D(t)-\phi_{MS}\frac{1}{2}\eta_S E_S^2(t)-\phi_{MP}\frac{1}{2}\eta_P E_P^2(t)-\left(1-\psi_R\right)\frac{1}{2}\eta_R E_R^2(t)\right]dt$$

The optimal value function of the retailer at t-time can be presented as:

$$\mathrm{V}_R(S,P)=\max_{E_R(t)}\int_t^{+\infty}e^{-\rho(\tau-t)}\left\{\pi_R D(t)-\phi_{MS}\frac{1}{2}\eta_S E_S^2(t)-\phi_{MP}\frac{1}{2}\eta_P E_P^2(t)-\left(1-\psi_R\right)\frac{1}{2}\eta_R E_R^2(t)\right\}dt \quad \text{EqD1}$$

For any S⩾0, P⩾0, $V_R(S,P)$ satisfies the HJB condition as follows:

$$\rho V_R(S,P)=\max_{E_R(t)}\left\{\pi_R(D_0+d_1S(t)+d_2P(t)+d_3E_R(t))-\phi_{MS}\frac{1}{2}\eta_S E_S^2(t)-\phi_{MP}\frac{1}{2}\eta_P E_P^2(t)-\left(1-\psi_R\right)\frac{1}{2}\eta_R E_R^2(t)+V_{RS}^{'}\left[s_1E_S(t)-\delta S(t)\right]+V_{RP}^{'}\left[p_1E_P(t)+p_2S(t)-\omega P(t)\right]\right\}dt \quad \text{EqD2}$$

The optimal value strategy satisfies that $\frac{\partial^2 \rho V_R(X,S)}{\partial E_R{}^2}=-\eta_P(1-\psi_R)<0$ and $\frac{\partial \rho V_R(X,S)}{\partial E_R}=\pi_R d_3-\eta_R(1-\psi_R)E_R(t)=0$, therefore, $E_R(t)=\frac{\pi_R d_3}{(1-\psi_R)\eta_R}$.

Let $E_R(t)$ in synergic digital supply chain with improved bilateral cost-sharing equal to $E_R(t)$ in collaboration-driven digital supply chain, ψ_R can be obtained as follow:

$$E_R^{SB*}(t)=E_R^{CD*}(t)\rightarrow\psi_R=1-\frac{\pi_R}{\pi_M+\pi_R}=\frac{\pi_M}{\pi_M+\pi_R} \quad \text{EqD3}$$

Then, the manufacturer make decision on $E_S(t),E_P(t)$ and ψ_R to maximize its own profit, and the optimal value function of manufacturer at t-moment can be presented as:

$$V_M(S,P)=\max_{E_S(t),E_P(t)}\int_t^{+\infty}e^{-\rho(\tau-t)}\left\{\pi_M D(t)-\left(1-\phi_{MS}\right)\frac{1}{2}\eta_S E_S^2(t)-\left(1-\phi_{MP}\right)\frac{1}{2}\eta_P E_P^2(t)-\psi_R\frac{1}{2}\eta_R E_R^2(t)+k(t)(S(t)-\overline{S})+k(t)(P(t)-\overline{P})\right\}dt \quad \text{EqD4}$$

Substitute $E_R(t)$, $E_P(t)$, and ψ_R into EqD4. For any S⩾0, P⩾0, $V_M(S,P)$ satisfies the HJB condition as follows:

$$\rho V_M(S,P)=\max_{E_S(t),E_P(t)}\left\{\begin{aligned}&\pi_M(D_0+d_1S(t)+d_2P(t)+\frac{\left(\pi_M+\pi_R\right)d_3{}^2}{\eta_R})-\left(1-\phi_{MS}\right)\frac{1}{2}\eta_S E_S^2(t)-\left(1-\phi_{MP}\right)\frac{1}{2}\eta_P E_P^2(t)-\frac{d_3{}^2\left(\pi_M+\pi_R\right)\pi_M}{2\eta_R}\\&+k(t)(S(t)-\overline{S})+k(t)(P(t)-\overline{P})+V_{MS}^{'}\left[s_1E_S(t)-\delta S(t)\right]+V_{MP}^{'}\left[p_1E_P(t)+p_2S(t)-\omega P(t)\right]\end{aligned}\right\}dt \quad \text{EqD5}$$

For $\frac{\partial \rho V_M(X,S)}{\partial E_S} = -\left(1-\phi_{MS}\right)\eta_S E_S(t) + s_1 V_{MS}^{'} = 0, \frac{\partial \rho V_M(X,S)}{\partial E_P} = -\left(1-\phi_{MP}\right)\eta_P E_P(t) + p_1 V_{MP}^{'} = 0$, we can obtain $E_S = \frac{s_1 V_{MS}^{'}}{\left(1-\phi_{MS}\right)\eta_S}$ and $E_P = \frac{p_1 V_{MP}^{'}}{\left(1-\phi_{MP}\right)\eta_P}$.

Let $E_S(t),\ E_P(t)$ in synergic digital supply chain with improved bilateral cost-sharing equal to $E_S(t),\ E_P(t)$ in collaboration-driven digital supply chain, φ_M can be obtained as follow:

$$\begin{cases} E_S^{SB*}(t) = E_S^{CD*}(t) \rightarrow \phi_{MS} = 1 - \frac{V_{MS}^{'}}{V_{TS}^{'\ CD}} = \frac{V_{TS}^{'\ CD} - V_{MS}^{'}}{V_{TS}^{'\ CD}} = \frac{(\rho+\omega)\pi_R d_1 + p_2 \pi_R d_2}{\left((\pi_M+\pi_R)d_1 + k(t)\right)(\rho+\omega) + p_2\left((\pi_M+\pi_R)d_2 + k(t)\right)} \\ E_P^{SB*}(t) = E_P^{CD*}(t) \rightarrow \phi_{MP} = 1 - \frac{V_{MP}^{'}}{V_{TP}^{'\ CD}} = \frac{V_{TP}^{'\ CD} - V_{MP}^{'}}{V_{TP}^{'\ CD}} = \frac{\pi_R d_2}{(\pi_M+\pi_R)d_2 + k(t)} \end{cases} \quad \text{EqD6}$$

Substitute $E_S(t)$ *and* $E_P(t)$ into EqD5 and simplify:

$$\rho \mathrm{V}_M(S,P) = \max_{E_S(t),E_P(t)} \begin{Bmatrix} \pi_M\left(D_0 + d_1 S(t) + d_2 P(t) + \frac{\left(\pi_M+\pi_R\right)d_3^{\ 2}}{\eta_R}\right) + \frac{s_1^{\ 2} V_{MS}^{'\ 2}}{2\left(1-\phi_{MS}\right)\eta_S} + \frac{p_1^{\ 2} V_{MP}^{'\ 2}}{2\left(1-\phi_{MP}\right)\eta_P} - \frac{d_3^{\ 2}\left(\pi_M+\pi_R\right)\pi_M}{2\eta_R} + k(t)(S(t)-\overline{S}) \\ + k(t)(P(t)-\overline{P}) - V_{MS}^{'}\delta S(t) + V_{MP}^{'} p_2 S(t) - V_{MP}^{'}\omega P(t) \end{Bmatrix} dt \quad \text{EqD7}$$

Since $\mathrm{V}_M(S,P)$ is the linear function of $S(t)$ and $P(t)$, we assume that $V_M(S,P) = a_1 S(t) + b_1 P(t) + c_1$, then $V_{MS}^{'} = a_1, V_{MP}^{'} = b_1$, where a_1, b_1, c_1 are constants.

$$\rho \mathrm{V}_M(S,P) = \left(\pi_M d_1 + k(t) - \delta V_{MS}^{'} + p_2 V_{MP}^{'}\right)S(t) + \left(\pi_M d_2 + k(t) - \omega V_{MP}^{'}\right)P(t) + \frac{\pi_M\left(\pi_M+\pi_R\right)d_3^{\ 2}}{2\eta_R} + \frac{s_1^{\ 2} V_{MS}^{'} V_{TS}^{'\ CD}}{2\eta_S} + \frac{p_1^{\ 2} V_{MP}^{'} V_{TP}^{'\ CD}}{2\eta_P} + \pi_M D_0 - k(t)\overline{S} - k(t)\overline{P} \quad \text{EqD8}$$

Substitute $V_M(S,P)$ and its first partial derivative into EqD8 and simplify, the following equation can be obtained:

$$\begin{cases} \rho a_1 = \pi_M d_1 + k(t) - \delta a_1 + p_2 b_1 \\ \rho b_1 = \pi_M d_2 + k(t) - \omega b_1 \\ \rho c_1 = \dfrac{\pi_M \left(\pi_M + \pi_R\right) d_3{}^2}{2\eta_R} + \dfrac{s_1{}^2 a_1 V_{TS}^{'\ CD}}{2\eta_S} + \dfrac{p_1{}^2 b_1 V_{TP}^{'\ CD}}{2\eta_P} + \pi_M D_0 - k(t)\overline{S} - k(t)\overline{P} \end{cases} \quad \text{EqD9}$$

the parameter values of a_1, b_1, c_1 can be solved as:

$$\begin{cases} a_1 = \dfrac{(\rho+\omega)\left(\pi_M d_1 + k(t)\right) + p_2\left(\pi_M d_2 + k(t)\right)}{(\rho+\delta)(\rho+\omega)} \\ b_1 = \dfrac{\pi_M d_2 + k(t)}{\rho+\omega} \\ c_1 = \dfrac{\pi_M \left(\pi_M + \pi_R\right) d_3{}^2}{2\eta_R \rho} + \dfrac{\pi_M D_0 - k(t)\overline{S} - k(t)\overline{P}}{\rho} \\ +s_1{}^2 \dfrac{\left((\rho+\omega)\left(\pi_M d_1 + k(t)\right) + p_2\left(\pi_M d_2 + k(t)\right)\right)}{2\eta_S \rho(\rho+\delta)^2(\rho+\omega)^2}\left(\left((\rho+\omega)\left(\pi_M d_1 + \pi_R d_1 + k(t)\right) + p_2\left(\pi_M d_2 + \pi_R d_2 + k(t)\right)\right)\right) \\ +p_1{}^2 \dfrac{\left(\pi_M d_2 + k(t)\right)}{2\eta_P \rho\left(\rho+\omega\right)^2}\left(\pi_M d_2 + \pi_R d_2 + k(t)\right) \end{cases} \quad \text{EqD10}$$

Therefore,

$$\begin{cases} E_S(t) = \dfrac{s_1 V_{MS}^{'}}{\left(1-\phi_{MS}\right)\eta_S} = \dfrac{s_1 V_{TS}^{'\ CD}}{\eta_S} = \dfrac{s_1((\pi_M + \pi_R)(d_1\rho + d_1\omega + d_2 p_2) + k(t)(\rho + \omega + p_2))}{\eta_S(\rho+\delta)(\rho+\omega)} \\ E_P(t) = \dfrac{p_1 V_{MP}^{'}}{(1-\phi_{MP})\eta_P} = \dfrac{p_1 V_{TP}^{'\ CD}}{\eta_P} = \dfrac{p_1((\pi_M + \pi_R)d_2 + k(t))}{\eta_P(\rho+\omega)} \\ E_R(t) = \dfrac{\pi_R d_3}{(1-\psi_R)\eta_R} = \dfrac{d_3(\pi_M + \pi_R)}{\eta_R} \end{cases} \quad \text{EqD11}$$

Substitute $E_S(t), E_P(t)$, $E_R(t)$ into EqD2 and simplify:

$$\rho \mathrm{V}_R(S,P)=\max_{E_S(t)}\left\{\pi_R(D_0+d_1S(t)+d_2P(t)+\frac{\left(\pi_M+\pi_R\right)d_3{}^2}{\eta_R})-\phi_{MS}\frac{1}{2}\eta_S E_S^2(t)-\phi_{MP}\frac{1}{2}\eta_P E_P^2(t)-\left(1-\psi_R\right)\frac{1}{2}\eta_R E_R^2(t)+V_{RS}^{'}\left[s_1E_S(t)-\delta S(t)\right]+V_{RP}^{'}\left[p_1E_P(t)+p_2S(t)-\omega P(t)\right]\right\}dt$$

EqD12

Since $\mathrm{V}_R(S,P)$ is the linear function of $S(t)$ and $P(t)$, we assume that $V_R(S,P)=a_2S(t)+b_2P(t)+c_2$, then $V_{RS}^{'}=a_2, V_{RP}^{'}=b_2$, where a_2,b_2,c_2 are constants.

$$\rho \mathrm{V}_R(S,P)=\left(\pi_R d_1-\delta V_{RS}^{'}+p_2V_{RP}^{'}\right)S(t)+\left(\pi_R d_2-\omega V_{RP}^{'}\right)P(t)+\frac{\pi_R\left(\pi_M+\pi_R\right)d_3{}^2}{2\eta_R}+\pi_R D_0+\frac{s_1{}^2V_{TS}^{'}\left(2V_{RS}^{'}-V_{TS}^{'}+V_{MS}^{'}\right)}{2\eta_S}+\frac{p_1{}^2V_{TP}^{'}\left(2V_{RP}^{'}-V_{TP}^{'}+V_{MP}^{'}\right)}{2\eta_P} \quad \text{EqD13}$$

Substitute $V_R(S,P)$ and its first partial derivative into EqD13 and simplify, the following equation can be obtained:

$$\begin{cases} \rho a_2=\pi_R d_1-\delta a_2+p_2b_2 \\ \rho b_2=\pi_R d_2-\omega b_2 \\ \rho c_2=\dfrac{\pi_R\left(\pi_M+\pi_R\right)d_3{}^2}{2\eta_R}+\pi_R D_0+\dfrac{s_1{}^2V_{TS}^{'}\left(2V_{RS}^{'}-V_{TS}^{'}+V_{MS}^{'}\right)}{2\eta_S}+\dfrac{p_1{}^2V_{TP}^{'}\left(2V_{RP}^{'}-V_{TP}^{'}+V_{MP}^{'}\right)}{2\eta_P} \end{cases} \quad \text{EqD14}$$

the parameter values of a_2,b_2,c_2 can be solved as:

$$\begin{cases} a_2 = \dfrac{\left(\rho+\omega\right)\pi_R d_1 + p_2\pi_R d_2}{(\rho+\delta)(\rho+\omega)} \\ b_2 = \dfrac{\pi_R d_2}{\rho+\omega} \\ c_2 = \dfrac{\pi_R\left(\pi_M+\pi_R\right){d_3}^2}{2\rho\eta_R} + \dfrac{\pi_R D_0}{\rho} + \dfrac{{s_1}^2 V_{TS}^{'}\left(2V_{RS}^{'} - V_{TS}^{'} + V_{MS}^{'}\right)}{2\rho\eta_S} + \dfrac{{p_1}^2 V_{TP}^{'}\left(2V_{RP}^{'} - V_{TP}^{'} + V_{MP}^{'}\right)}{2\rho\eta_P} \\ = \dfrac{\pi_R\left(\pi_M+\pi_R\right){d_3}^2}{2\rho\eta_R} + \dfrac{\pi_R D_0}{\rho} + {s_1}^2\dfrac{\left((\rho+\omega)(d_1\pi_M + d_1\pi_R + k(t)) + p_2(d_2\pi_M + d_2\pi_R + k(t))\right)}{2\rho\eta_S(\rho+\delta)^2(\rho+\omega)^2}\left(\left(\rho+\omega\right)\pi_R d_1 + p_2\pi_R d_2\right) + \dfrac{{p_1}^2\left((\pi_M+\pi_R)d_2 + k(t)\right)\pi_R d_2}{2\rho\eta_P\left(\rho+\omega\right)^2} \end{cases} \quad \text{EqD15}$$

Due to the optimal strategies of the retailer, the manufacturer under improved bilateral cost-sharing equal the optimal strategies in collaboration-driven digital supply chain, the optimal strategy of the government and the state of trajectory of the carbon adatement level and the digital level and under improved bilateral cost-sharing equal the those in collaboration-driven digital supply chain.

Appendix E Proof of Corollary1

Intuitively, $k^{CD*}(t)=k^{MD*}(t)$, By comparing EqA19 to EqB21, we can get $E_S^{CD^*}>E_S^{MD^*}, E_P^{CD^*}>E_P^{MD^*}, E_R^{CD^*}>E_R^{MD^*}$

By comparing EqA20 to EqB22, we can get $S_\infty^{CD}>S_\infty^{MD}$

$S^{CD*}(t)-S^{MD*}(t)=S_\infty^{CD}+(S_0-S_\infty^{CD})e^{-\delta t}-S_\infty^{MD}-(S_0-S_\infty^{MD})e^{-\delta t}=\left(S_\infty^{CD}-S_\infty^{MD}\right)\left(1-e^{-\delta t}\right)$, therefore, $S^{CD*}(t)>S^{MD*}(t)$

By comparing EqA23 and EqB25, we can get $\frac{P_\infty^{CD}}{P_\infty^{MD}}>1$

$$P^{CD*}(t)-P^{MD*}(t)=P_\infty^{CD}+\left(P_0-P_\infty^{CD}-\frac{p_2(S_0-S_\infty^{CD})}{\omega-\delta}\right)e^{-\omega t}+\frac{p_2(S_0-S_\infty^{CD})}{\omega-\delta}e^{-\delta t}-P_\infty^{MD}-\left(P_0-P_\infty^{MD}-\frac{p_2(S_0-S_\infty^{MD})}{\omega-\delta}\right)e^{-\omega t}-\frac{p_2(S_0-S_\infty^{MD})}{\omega-\delta}e^{-\delta t}$$
$$=P_\infty^{CD}-P_\infty^{MD}+\left(\frac{p_2(S_0-S_\infty^{CD})}{\omega-\delta}-P_\infty^{CD}\right)e^{-\omega t}+\frac{p_2(S_0-S_\infty^{CD})}{\omega-\delta}e^{-\delta t}-P_\infty^{MD}-\left(P_0-P_\infty^{MD}-\frac{p_2(S_0-S_\infty^{MD})}{\omega-\delta}\right)e^{-\omega t}-\frac{p_2(S_0-S_\infty^{MD})}{\omega-\delta}e^{-\delta t}$$
$$=\left(P_\infty^{CD}-P_\infty^{MD}\right)\left(1-e^{-\omega t}\right)+\frac{p_2(S_\infty^{MD}-S_\infty^{CD})}{\omega-\delta}\left(e^{-\omega t}-e^{-\delta t}\right)$$

If $\omega>\delta$, then $\frac{p_2(S_\infty^{MD}-S_\infty^{CD})}{\omega-\delta}<0, e^{-\omega t}-e^{-\delta t}<0$ and $\frac{p_2(S_\infty^{MD}-S_\infty^{CD})}{\omega-\delta}\left(e^{-\omega t}-e^{-\delta t}\right)>0$.

If $\omega<\delta$, then $\frac{p_2(S_\infty^{MD}-S_\infty^{CD})}{\omega-\delta}>0, e^{-\omega t}-e^{-\delta t}>0$ and $\frac{p_2(S_\infty^{MD}-S_\infty^{CD})}{\omega-\delta}\left(e^{-\omega t}-e^{-\delta t}\right)>0$

Since $\left(P_\infty^{CD}-P_\infty^{MD}\right)\left(1-e^{-\omega t}\right)>0$, we can conclude that $P^{CD*}(t)-P^{MD*}(t)>0$, $P^{CD*}(t)>P^{MD*}(t)$ and $D^{CD*}(t)>D^{MD*}(t)$..

By comparing EqA9 to EqB8 and EqB11, we can obtain $a_1^{MD*}+a_2^{MD*}<a_1^{CD*}$, $b_1^{MD*}+b_2^{MD*}<b_1^{CD*}$, $c_1^{MD*}+c_2^{MD*}=c_1^{CD*}$, then $R_T^{CD*}>R_T^{MD*}$.

Appendix F Proof of Corollary2

Comparing EqA19 and EqC22, we can obtain that $k^{SU*}(t)=k^{CD*}(t),\quad E_S^{CD^*}>E_S^{SU^*},E_P^{CD^*}>E_P^{SU^*},E_R^{CD^*}>E_R^{SU^*}$

By comparing EqA21 and EqC24, we can get $S_\infty^{CD}>S_\infty^{SU}$

$S^{CD*}(t)-S^{SU*}(t)=S_\infty^{CD}+(S_0-S_\infty^{SU})e^{-\delta t}-S_\infty^{SU}-(S_0-S_\infty^{SU})e^{-\delta t}=\left(S_\infty^{CD}-S_\infty^{SU}\right)\left(1-e^{-\delta t}\right)$, therefore, $S^{CD*}(t)>S^{SU*}(t)$

By comparing EqA23 and EqC26, we can get $\dfrac{P_\infty^{CD}}{P_\infty^{SU}}>1$

$$P^{CD*}(t)-P^{SU*}(t)=P_\infty^{CD}+\left(P_0-P_\infty^{CD}-\frac{p_2(S_0-S_\infty^{CD})}{\omega-\delta}\right)e^{-\omega t}+\frac{p_2(S_0-S_\infty^{CD})}{\omega-\delta}e^{-\delta t}-P_\infty^{SU}-\left(P_0-P_\infty^{SU}-\frac{p_2(S_0-S_\infty^{SU})}{\omega-\delta}\right)e^{-\omega t}-\frac{p_2(S_0-S_\infty^{SU})}{\omega-\delta}e^{-\delta t}$$
$$=P_\infty^{CD}-P_\infty^{SU}+\left(\frac{p_2(S_0-S_\infty^{CD})}{\omega-\delta}-P_\infty^{CD}\right)e^{-\omega t}+\frac{p_2(S_0-S_\infty^{CD})}{\omega-\delta}e^{-\delta t}-P_\infty^{SU}-\left(P_0-P_\infty^{SU}-\frac{p_2(S_0-S_\infty^{SU})}{\omega-\delta}\right)e^{-\omega t}-\frac{p_2(S_0-S_\infty^{SU})}{\omega-\delta}e^{-\delta t}$$
$$=\left(P_\infty^{CD}-P_\infty^{SU}\right)\left(1-e^{-\omega t}\right)+\frac{p_2(S_\infty^{SU}-S_\infty^{CD})}{\omega-\delta}\left(e^{-\omega t}-e^{-\delta t}\right)$$

If $\omega>\delta$, then $\dfrac{p_2(S_\infty^{SU}-S_\infty^{CD})}{\omega-\delta}<0, e^{-\omega t}-e^{-\delta t}<0$ and $\dfrac{p_2(S_\infty^{SU}-S_\infty^{CD})}{\omega-\delta}\left(e^{-\omega t}-e^{-\delta t}\right)>0$.

If $\omega<\delta$, then $\dfrac{p_2(S_\infty^{SU}-S_\infty^{CD})}{\omega-\delta}>0, e^{-\omega t}-e^{-\delta t}>0$ and $\dfrac{p_2(S_\infty^{SU}-S_\infty^{CD})}{\omega-\delta}\left(e^{-\omega t}-e^{-\delta t}\right)>0$

Since $\left(P_\infty^{CD}-P_\infty^{SU}\right)\left(1-e^{-\omega t}\right)>0$, we can conclude that $P^{CD*}(t)-P^{SU*}(t)>0,\ P^{CD*}(t)>P^{SU*}(t)$ and $D^{CD*}(t)>D^{SU*}(t)$.

By comparing EqA9 to EqC9 and EqC13, we can obtain $a_1^{SU*}+a_2^{SU*}<a_1^{CD*},\ b_1^{SU*}+b_2^{SU*}<b_1^{CD*},\ c_1^{SU*}+c_2^{SU*}=c_1^{CD*}$, then $\pi_T^{CD*}>\pi_T^{SU*}$.

Next, we compare EqB21 and EqC22, we can obtain that $k^{SU*}(t)=k^{MD*}(t)$, $E_S^{SU^*}=E_S^{MD^*}, E_P^{SU^*}=E_P^{MD^*}$, and $E_R^{SU^*}>E_R^{MD^*}$ only if $2\pi_M-\pi_R>0$ is hold.

By comparing EqB23 and EqC24, we can get $S_\infty^{SU}=S_\infty^{MD}$.

$S^{SU*}(t)-S^{MD*}(t)=S_\infty^{SU}+(S_0-S_\infty^{SU})e^{-\delta t}-S_\infty^{MD}-(S_0-S_\infty^{MD})e^{-\delta t}=\left(S_\infty^{SU}-S_\infty^{MD}\right)\left(1-e^{-\delta t}\right)$, therefore, $S^{SU*}(t)=S^{MD*}(t)$.

By comparing EqB25 and EqC26, we can get $\frac{P_\infty^{SU}}{P_\infty^{MD}}=1$.

$$P^{MD*}(t)-P^{SU*}(t)=P_\infty^{MD}+\left(P_0-P_\infty^{MD}-\frac{p_2(S_0-S_\infty^{MD})}{\omega-\delta}\right)e^{-\omega t}+\frac{p_2(S_0-S_\infty^{MD})}{\omega-\delta}e^{-\delta t}-P_\infty^{SU}-\left(P_0-P_\infty^{SU}-\frac{p_2(S_0-S_\infty^{SU})}{\omega-\delta}\right)e^{-\omega t}-\frac{p_2(S_0-S_\infty^{SU})}{\omega-\delta}e^{-\delta t}$$
$$=P_\infty^{MD}-P_\infty^{SU}+\left(\frac{p_2(S_0-S_\infty^{MD})}{\omega-\delta}-P_\infty^{MD}\right)e^{-\omega t}+\frac{p_2(S_0-S_\infty^{MD})}{\omega-\delta}e^{-\delta t}-P_\infty^{SU}-\left(P_0-P_\infty^{SU}-\frac{p_2(S_0-S_\infty^{SU})}{\omega-\delta}\right)e^{-\omega t}-\frac{p_2(S_0-S_\infty^{SU})}{\omega-\delta}e^{-\delta t}$$
$$=\left(P_\infty^{MD}-P_\infty^{SU}\right)\left(1-e^{-\omega t}\right)+\frac{p_2(S_\infty^{SU}-S_\infty^{MD})}{\omega-\delta}\left(e^{-\omega t}-e^{-\delta t}\right)$$

Since $P_\infty^{MD}-P_\infty^{SU}=0$, we can conclude that $P^{MD*}(t)=P^{SU*}(t)$ and $D^{MD*}(t)<D^{SU*}(t)$.

By comparing the sum of EqB8 to EqB11 with the sum of EqC9 and EqC13, we can obtain the following results:

$a_1^{SU*}+a_2^{SU*}=a_1^{MD*}+a_2^{MD*}$, $b_1^{SU*}+b_2^{SU*}=b_1^{MD*}+b_2^{MD*}$ and $c_1^{SU*}+c_2^{SU*}>c_1^{MD*}+c_2^{MD*}$ with the condition $2\pi_M-\pi_R>0$, then $R_T^{SU*}>R_T^{MD*}$ when $2\pi_M-\pi_R>0$

$a_1^{SU*}=a_1^{MD*}$, $b_1^{SU*}=b_1^{MD*}$ and $c_1^{SU*}>c_1^{MD*}$ with the condition $2\pi_M-\pi_R>0$, then $R_M^{SU*}>R_M^{MD*}$ when $2\pi_M-\pi_R>0$
$a_2^{SU*}=a_2^{MD*}$, $b_2^{SU*}=b_2^{MD*}$ and $c_2^{SU*}>c_2^{MD*}$ with the condition $2\pi_M-\pi_R>0$, then $R_R^{SU*}>R_R^{MD*}$ when $2\pi_M-\pi_R>0$

Taking Corollary 1 into consideration, we can obtain the following corollary when $2\pi_M>\pi_R$

$$k^{SU*}(t) = k^{CD*}(t) = k^{MD*}(t), E_S^{MD^*} = E_S^{SU^*} < E_S^{CD^*}, E_P^{MD^*} = E_P^{SU^*} < E_P^{CD^*}, E_R^{MD^*} < E_R^{SU^*} < E_R^{CD^*}\ , D^{MD^*} < D^{SU^*} < D^{CD^*}$$

$$S_\infty^{MD} = S_\infty^{SU} < S_\infty^{CD}, P_\infty^{MD} = P_\infty^{SU} < P_\infty^{CD}, R_T^{MD*} < R_T^{SU*} < R_T^{CD*}$$

Appendix G Proof of Corollary3

In the Appendix D, we have proved that $k^{SB*}(t) = k^{CD*}(t),\quad E_S^{SB*} = E_S^{CD*}, E_P^{SB*} = E_P^{CD*}, E_R^{SB*} = E_R^{CD*}, S_\infty^{SB} = S_\infty^{CD}, P_\infty^{SB} = P_\infty^{SU}$

By comparing the sum of EqD10 and EqD15 to EqA9, we can obtain $a_1^{SB*} + a_2^{SB*} = a_1^{CD*},\ b_1^{SB*} + b_2^{SB*} = b_1^{SB*},\ c_1^{SB*} + c_2^{SB*} = c_1^{CB*}$, then $\pi_T^{CD*} > \pi_T^{SU*}$.

Taking Corollary 1 into consideration, we can obtain the following corollary:

$k^{SB*}(t) = k^{CD*}(t) > k^{MD*}(t),\ E_S^{SB*} = E_S^{CD*} > E_S^{MD^*},\ E_P^{SB*} = E_P^{CD*} > E_P^{MD^*},\ E_R^{SB*} = E_R^{CD*} > E_R^{MD^*}$

$S_\infty^{SB} = S_\infty^{CD} > S_\infty^{MD},\ P_\infty^{SB} = P_\infty^{SU} > P_\infty^{CD},\ R_T^{SB*} = R_T^{CD*} > R_T^{MD*}$

Appendix H Proof of Corollary4

Setting appropriate profit sharing ratios in a collaborative digital supply chain with bilateral participation, the profits of the manufacturer and the retailer respectively are $R_M^{SB'} = \lambda R_T^{SB*} = \lambda R_T^{CD*}$ and $R_R^{SB'} = (1-\lambda) R_T^{SB*} = (1-\lambda) R_T^{CD*}$ respectively.

To ensure that the manufacturer and retailer receive Pareto improvements relative to the market-driven scenario in the bilateral participation, this condition need to be satisfied: $R_M^{SB'} = \lambda R_T^{SB*} = \lambda R_T^{CD*} > R_M^{MD*}$ and $R_R^{SB'} = (1-\lambda) R_T^{SB*} = (1-\lambda) R_T^{CD*} > R_R^{MD*}$.

There is a range for this profit sharing ratio $\frac{R_M^{MD*}}{R_T^{CD*}} < \lambda < \frac{R_T^{CD*} - R_R^{MD*}}{R_T^{CD*}}$, but λ is ultimately determined by the bargaining power of the manufacturer and the retailer. According to Rubinstein's bargaining model, the optimal equilibrium of the share of the manufacturer's total profit in a synergistic supply chain with bilateral participation is $\lambda = \frac{1-e_R}{1-e_R e_M}\left(\frac{R_T^{CD*} - R_R^{MD*}}{R_T^{CD*}} - \frac{R_M^{MD*}}{R_T^{CD*}}\right) + \frac{R_M^{MD*}}{R_T^{CD*}}$, where e_R and e_M are the discount factor, also called patience, are determined by the affordability and psychological capacity of the manufacturer and the retailer.